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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

# Solutions

JEE Main Chemistry · Previous Year Questions with Answer Key

| 208 | 179 | 2002 – 2026 | English |
|---|---|---|---|
| QUESTIONS | PAPERS COVERED | YEARS | LANGUAGE |

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## MCQ (Single Correct Answer) 123 questions

**Q1.** JEE Main 2026 (Online) 8th April Evening Shift

Open solution

Given below are two statements :

Given : Molar mass of $C, H, O, Cl$ are $12, 1, 16$ and $35.5\ g\ mol^{-1}$, respectively

**Statement I :** In $30\%(w/w)$ solution of methanol in $CCl_4$( at T K), the mole fraction of $CCl_4$ is equal to 0.33 .

**Statement II :** Mixture of methanol and $CCl_4$ shows positive deviation from Raoult's law. In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

**Q2.** JEE Main 2026 (Online) 8th April Evening Shift

Open solution

Which of the following statements are not correct?

A. For water, magnitude of $K_b$ is more than the magnitude of $K_f$.

B. The elevation in boiling point of water when a non-volatile solute is added to it is larger in magnitude than its depression in freezing point.

C. Osmotic pressure measurement is preferred over any other colligative property to determine molar mass of proteins and polymers.

D. The dimerised form of benzoic acid in benzene is



Choose the correct answer from the options given below :

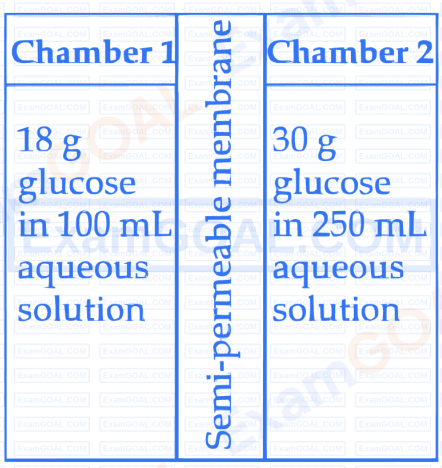
(A) A and B only (B) A and D only (C) A, B and D only (D) A, C and D only

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**Q3.** JEE Main 2026 (Online) 6th April Evening Shift

Open solution

Given below are two statements :




**Statement I :** $H_2O$ molecules move from the chamber 1 to chamber 2 .

**Statement II :** The osmotic pressure of a solution prepared by dissolving 50 mg of potassium sulphate (molar mass $= 174\ g/mol$ ) in 2 L of water (at $27^\circ C$ ) is 0.0107 bar. (Given: $R = 0.083 dm^3$ bar $K^{-1}\ mol^{-1}$ and assume complete dissociation of electrolyte)

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

**Q4.** JEE Main 2026 (Online) 6th April Morning Shift

Open solution

When 0.25 moles of a non-volatile, non-ionizable solute was dissolved in 1 mole of a solvent the vapor pressure of solution was $x\%$ of vapor pressure of pure solvent. What is $x\%$ ?

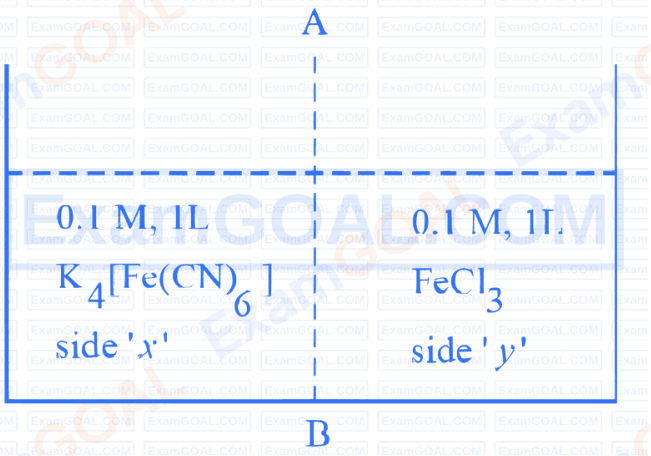
(A) 50% (B) 60% (C) 70% (D) 80%

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**Q5.** JEE Main 2026 (Online) 4th April Evening Shift

Open solution

At $27^{\circ}C, 0.1M, 1\ L\ K_4\ [Fe(CN)_6]$ aqueous solution and $0.1M, 1\ L FeCl_3$ aqueous solution are placed in a container separated by a semi permeable membrane AB . Assume complete dissociation of both the solutes. Which of the following statement is correct?




(A) Blue color is formed on both sides.

(B) Ionic solutes in aqueous solution can pass through semi-permeable membrane.

(C) Solution on side ' y ' is hypotonic.

(D) To cause the reverse flow of solvent during osmosis, external pressure (any value) should be applied to side ' $x$ '.

**Q6.** JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Solution A is prepared by dissolving 1 g of a protein (molar mass = 50,000 g $mol^{-1}$) in 0.5 L of water at 300 K. Its osmotic pressure is $x$ bar. Solution B is made by dissolving 2 g of the same protein in 1 L of water at 300 K. Osmotic pressure of solution B is $y$ bar. Entire solution of A is mixed with entire solution of B at the same temperature. The osmotic pressure of resultant solution is $z$ bar. $x$, $y$ and $z$ respectively are:

(R = 0.083 L bar $mol^{-1}$ $K^{-1}$)

(A) $9.96 \times 10^{-4}$; $9.96 \times 10^{-4}$; $9.96 \times 10^{-4}$

(B) $9.96 \times 10^{-4}$; $9.96 \times 10^{-4}$; $19.92 \times 10^{-4}$

(C) $4.98 \times 10^{-4}$; $4.98 \times 10^{-4}$; $9.96 \times 10^{-4}$

(D) $4.98 \times 10^{-4}$; $4.98 \times 10^{-4}$; $4.98 \times 10^{-4}$

**Q7.** JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

19.5 g of fluoro acetic acid (molar mass = 78 g $mol^{-1}$) is dissolved in 500 g of water at 298 K. The depression in the freezing point of water was 1 °C. What is $K_a$ of fluoro acetic acid?

(For water, $K_f = 1.86$ K kg $mol^{-1}$). Assume molarity and molality to have same values.

(A) $10^{-6}$

(B) $4 \times 10^{-4}$

(C) $3 \times 10^{-5}$

(D) $3 \times 10^{-3}$

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**Q8.** JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Consider the following aqueous solutions.

I. 2.2 g Glucose in 125 mL of solution.

II. 1.9 g Calcium chloride in 250 mL of solution.

III. 9.0 g Urea in 500 mL of solution.

IV. 20.5 g Aluminium sulphate in 750 mL of solution.

The **correct** increasing order of boiling point of these solutions will be :

[Given : Molar mass in g $mol^{-1}$ : H = 1, C = 12, N = 14, O = 16, Cl = 35.5, Ca = 40, Al = 27 and S = 32]

(A) III < I < II < IV

(B) II < III < IV < I

(C) II < III < I < IV

(D) I < II < III < IV

**Q9.** JEE Main 2026 (Online) 28th January Morning Shift

Open solution

At $\mathrm{T(K)}$, 2 moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution formed is 320 mm Hg . At this stage, one mole of A and one mole of B are added to the solution. The vapour pressure is now measured as 328.6 mm Hg . The vapour pressure (in mm Hg ) of A and B are respectively:

(A) 400, 300

(B) 600, 400

(C) 500, 200

(D) 300, 200

**Q10.** JEE Main 2026 (Online) 24th January Evening Shift

Open solution

At 298 K , the mole percentage of $\mathrm{N_2(g)}$ in air is $80\%$. Water is in equilibrium with air at a pressure of 10 atm . What is the mole fraction of $\mathrm{N_2(g)}$ in water at 298 K ? ($\mathrm{K_H}$ for $\mathrm{N_2}$ is $6.5 \times 10^7$ mmHg)

(A) $9.35 \times 10^{-5}$

(B) $9.35 \times 10^{5}$

(C) $1.23 \times 10^{-7}$

(D) $1.17 \times 10^{-4}$

**Q11.** JEE Main 2026 (Online) 24th January Evening Shift

Open solution

Two liquids A and B form an ideal solution at temperature T K . At T K , the vapour pressures of pure A and B are 55 and $15 \mathrm{kNm^{-2}}$ respectively. What is the mole fraction of A in solution of A and B in equilibrium with a vapour in which the mole fraction of A is 0.8?

(A) 0.5217

(B) 0.480

(C) 0.340

(D) 0.663

**Q12.** JEE Main 2026 (Online) 24th January Morning Shift

Open solution

A solution is prepared by dissolving 0.3 g of a non-volatile non-electrolyte solute 'A' of molar mass $60\ \mathrm{g\ mol^{-1}}$ and 0.9 g of a non-volatile non-electrolyte solute ' B ' of molar mass $180\ \mathrm{g\ mol^{-1}}$ in $100\ \mathrm{mL H_2O}$ at $27^{\circ}\mathrm{C}$. Osmotic pressure of the solution will be

[Given: $\mathrm{R} = 0.082\ \mathrm{L\ atm\ K^{-1}\ mol^{-1}}$ ]

(A) 2.46 atm

(B) 0.82 atm

(C) 1.47 atm

(D) 1.23 atm

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**Q13.** JEE Main 2026 (Online) 24th January Morning Shift

Open solution

' W ' g of a non-volatile electrolyte solid solute of molar mass ' M ' $\mathrm{gmol}^{-1}$ when dissolved in 100 mL water, decreases vapour pressure of water from 640 mm Hg to 600 mm Hg . If aqueous solution of the electrolyte boils at 375 K and $\mathrm{K_b}$ for water is $0.52\ \mathrm{K\ kg\ mol}^{-1}$, then the mole fraction of the electrolyte solute $(x_2)$ in the solution can be expressed as

(Given : density of water $= 1\ \mathrm{g/mL}$ and boiling point of water $= 373\ \mathrm{K}$ )

(A) $\frac{16}{2.6} \times \frac{W}{M}$

(B) $\frac{1.3}{8} \times \frac{\mathrm{M}}{\mathrm{W}}$

(C) $\frac{2.6}{16} \times \frac{\mathrm{M}}{\mathrm{W}}$

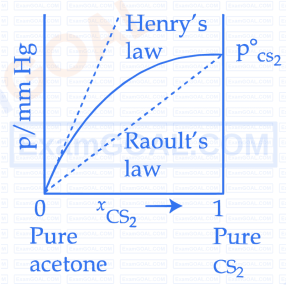
(D) $\frac{1.3}{8} \times \frac{W}{M}$

**Q14.** JEE Main 2026 (Online) 23rd January Morning Shift

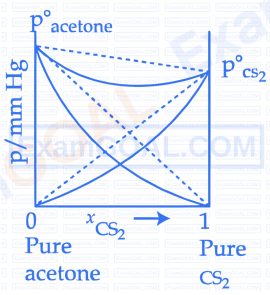
Open solution

Which one of the following graphs accurately represents the plot of partial pressure of $\mathrm{CS_2}$ vs its mole fraction in a mixture of acetone and $\mathrm{CS_2}$ at constant temperature?

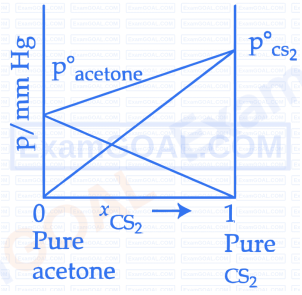
(A)



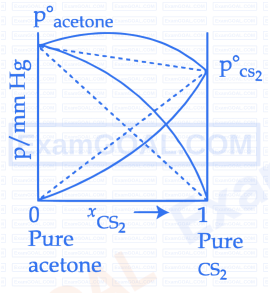

(B)




(C)




(D)




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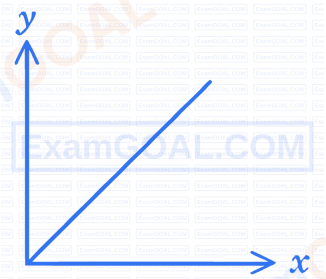
**Q15.** JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

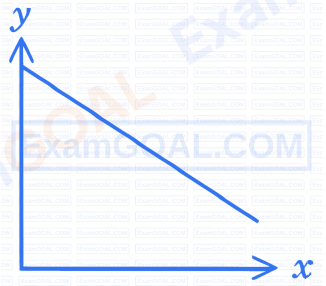
Consider a solution of $CO_2(g)$ dissolved in water in a closed container.

Which one of the following plots correctly represents variation of log (partial pressure of $CO_2$ in vapour phase above water) [$y$-axis ] with log (mole fraction of $CO_2$ in water) [$x$-axis ] at $25^\circ C$ ?

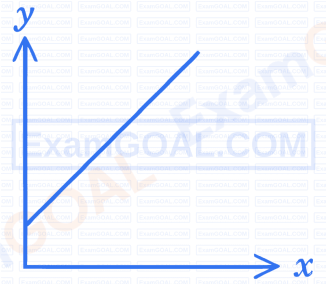
(A)



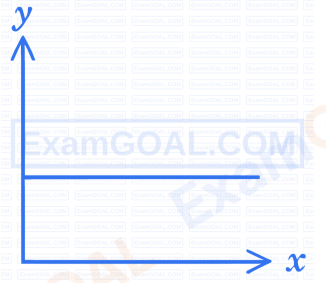

(B)




(C)




(D)




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**Q16.** JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Given below are two statements :

**Statement I :** The Henry's law constant $K_H$ is constant with respect to variations in solution's concentration over the range for which the solution is ideally dilute.

**Statement II :** $K_H$ does not differ for the same solute in different solvents.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

**Q17.** JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Elements P and Q form two types of non-volatile, non-ionizable compounds PQ and $PQ_2$. When 1 g of $PQ$ is dissolved in 50 g of solvent ' $A'$, $\Delta T_b$ was 1.176 K while when 1 g of $PQ_2$ is dissolved in 50 g of solvent ' A', $\Delta T_b$ was 0.689 K . ( $K_b$ of ' $A' = 5\ K\ kg\ mol^{-1}$ ). The molar masses of elements P and Q (in $gmol^{-1}$ ) respectively, are :

(A) 25, 60 (B) 60, 25 (C) 65, 145 (D) 70, 110

**Q18.** JEE Main 2025 (Online) 8th April Evening Shift

Open solution

$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$

The freezing point depression of a 0.1 m aqueous solution of a monobasic weak acid HA is 0.20 °C. The dissociation constant for the acid is

Given: $K_f(H_2O)$ = 1.8 K kg mol$^{-1}$, molality ≡ molarity

(A) $1.90 \times 10^{-3}$ (B) $1.38 \times 10^{-3}$ (C) $1.1 \times 10^{-2}$ (D) $1.89 \times 10^{-1}$

**Q19.** JEE Main 2025 (Online) 8th April Evening Shift

Open solution

Which of the following binary mixture does not show the behaviour of minimum boiling azeotropes?

(A) $CH_3OH + CHCl_3$ (B) $C_6H_5OH + C_6H_5NH_2$

(C) $H_2O + CH_3COC_2H_5$ (D) $CS_2 + CH_3COCH_3$

**Q20.** JEE Main 2025 (Online) 7th April Evening Shift

Open solution

Liquid A and B form an ideal solution. The vapour pressures of pure liquids A and B are 350 and 750 mm Hg respectively at the same temperature. If $x_A$ and $x_B$ are the mole fraction of A and B in solution while $y_A$ and $y_B$ are the mole fraction of A and B in vapour phase, then,

(A) $(x_A - y_A) < (x_B - y_B)$ (B) $\frac{x_A}{x_B} = \frac{y_A}{y_B}$

(C) $\frac{x_A}{x_B} < \frac{y_A}{y_B}$ (D) $\frac{x_A}{x_B} > \frac{y_A}{y_B}$

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**Q21.** JEE Main 2025 (Online) 7th April Evening Shift

Open solution

Match **List - I** with **List - II**.

| List - I | List - II |
|---|---|
| (A) Solution of chloroform and acetone | (I) Minimum boiling azeotrope |
| (B) Solution of ethanol and water | (II) Dimerizes |
| (C) Solution of benzene and toluene | (III) Maximum boiling azeotrope |
| (D) Solution of acetic acid in benzene | (IV) $\Delta V_{mix} = 0$ |

**Choose the correct answer from the options given below :**

**(A)** (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

**(B)** (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

**(C)** (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

**(D)** (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

**Q22.** JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Given below are two statements:

Statement (I) : Molal depression constant $\mathrm{K}_f$ is given by $\frac{\mathrm{M}_1\mathrm{RT}_f}{\Delta \mathrm{S}_{\mathrm{fus}}}$, where symbols have their usual meaning.

Statement (II) : $\mathrm{K}_f$ for benzene is less than the $\mathrm{K}_f$ for water.

In the light of the above statements, choose the most appropriate answer from the options given below :

**(A)** Statement I is correct but Statement II is incorrect

**(B)** Both Statement I and Statement II are incorrect

**(C)** Statement I is incorrect but Statement II is correct

**(D)** Both Statement I and Statement II are correct

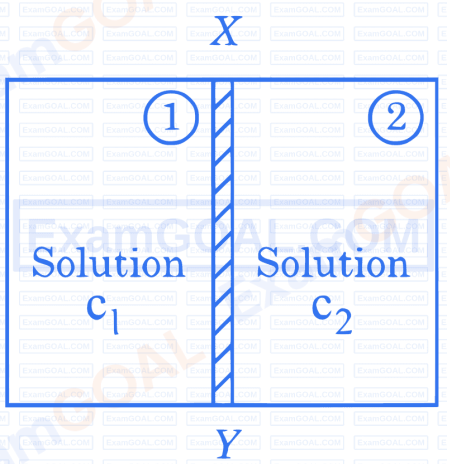
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**Q23.** JEE Main 2025 (Online) 4th April Morning Shift

Open solution

$XY$ is the membrane/partition between two chambers 1 and 2 containing sugar solutions of concentration $c_1$ and $c_2$ $(c_1 > c_2)\text{molL}^{-1}$. For the reverse osmosis to take place identify the correct condition.

(Here $p_1$ and $p_2$ are pressures applied on chamber 1 and 2 ).




A. Membrane/Partition : Cellophane, $\text{p}_1 > \pi$

B. Membrane/Partition : Porous, $\text{p}_2 > \pi$

C. Membrane/Partition : Parchment paper, $p_1 > \pi$

D. Membrane/Partition : Cellophane, $\text{p}_2 > \pi$

Choose the correct answer from the option given below:

**(A)** A and C Only **(B)** A and D Only **(C)** B and D Only **(D)** C Only

**Q24.** JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

2 moles each of ethylene glycol and glucose are dissolved in 500 g of water. The boiling point of the resulting solution is: (Given : Ebullioscopic constant of water $= 0.52\ \text{K kg mol}^{-1}$ )

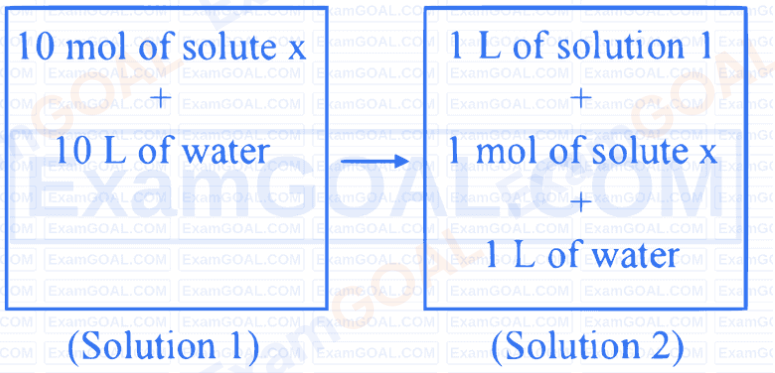
**(A)** 377.3 K **(B)** 379.2 K **(C)** 375.3 K **(D)** 277.3 K

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**Q25.** JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

Which of the following properties will change when system containing solution 1 will become solution 2 ?




(A) Molar heat capacity

(B) Concentration

(C) Gibbs free energy

(D) Density

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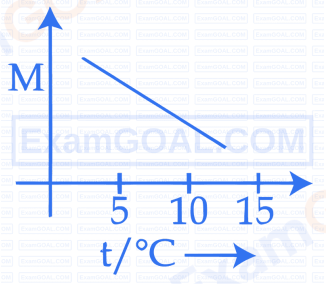
**Q26.** JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

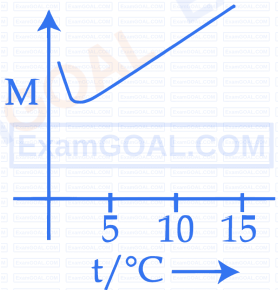
' $x$ ' g of NaCl is added to water in a beaker with a lid. The temperature of the system is raised from $1^{\circ}\mathrm{C}$ to $25^{\circ}\mathrm{C}$. Which out of the following plots, is best suited for the change in the molarity $(\mathrm{M})$ of the solution with respect to temperature ?

[Consider the solubility of NaCl remains unchanged over the temperature range]

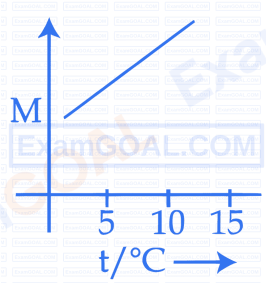
(A)



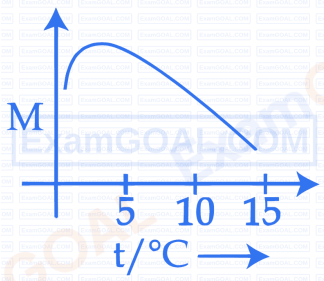

(B)




(C)




(D)




**Q27.** JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

A solution is made by mixing one mole of volatile liquid $A$ with 3 moles of volatile liquid $B$. The vapour pressure of pure A is 200 mm Hg and that of the solution is 500 mm Hg . The vapour pressure of pure B and the least volatile component of the solution, respectively, are:

(A) $1400\ \mathrm{mm\ Hg}$, A

(B) $1400\ \mathrm{mm\ Hg}$, $B$

(C) $600\ \mathrm{mm\ Hg}$, A

(D) $600\ \mathrm{mm\ Hg}$, B

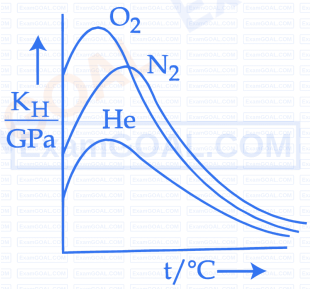
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**Q28.** JEE Main 2025 (Online) 2nd April Morning Shift

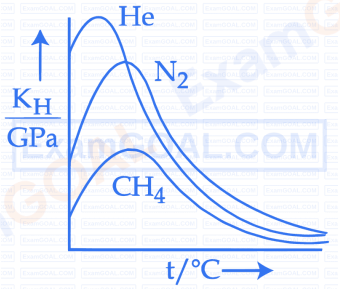
Open solution

Which of the following graph correctly represents the plots of $K_H$ at 1 bar for gases in water versus temperature?

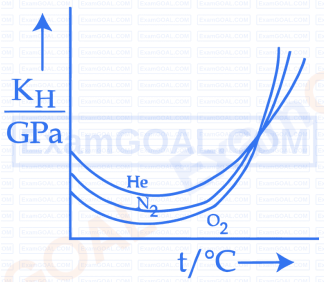
(A)



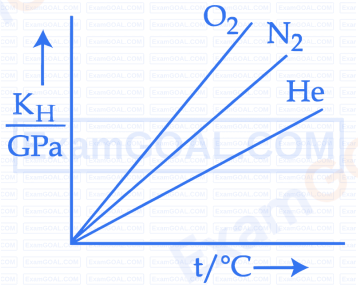

(B)




(C)




(D)




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**Q29.** JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Given below are two statements :

**Statement (I):** NaCl is added to the ice at 0°C, present in the ice cream box to prevent the melting of ice cream.

**Statement (II):** On addition of NaCl to ice at 0°C, there is a depression in freezing point.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

**Q30.** JEE Main 2025 (Online) 29th January Morning Shift

Open solution

1.24 g of $AX_2$ (molar mass 124 g $mol^{-1}$) is dissolved in 1 kg of water to form a solution with boiling point of 100.0156°C, while 25.4 g of $AY_2$ (molar mass 250 g $mol^{-1}$) in 2 kg of water constitutes a solution with a boiling point of 100.0260°C.

$K_b(H_2O)$ = 0.52 kg $mol^{-1}$

Which of the following is correct?

(A) $AX_2$ and $AY_2$ (both) are completely unionised.

(B) $AX_2$ and $AY_2$ (both) are fully ionised.

(C) $AX_2$ is completely unionised while $AY_2$ is fully ionised.

(D) $AX_2$ is fully ionised while $AY_2$ is completely unionised.

**Q31.** JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Assume a living cell with 0.9% (w/w) of glucose solution (aqueous). This cell is immersed in another solution having equal mole fraction of glucose and water.
(Consider the data upto first decimal place only)

The cell will :

(A) shrink since solution is $0.45\%(w/w)$ as a result of association of glucose molecules (due to hydrogen bonding)

(B) shrink since solution is $0.5\%(w/w)$

(C) show no change in volume since solution is $0.9\%(w/w)$

(D) swell up since solution is $1\%(w/w)$

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**Q32.** JEE Main 2025 (Online) 28th January Morning Shift

Open solution

What is the freezing point depression constant of a solvent, 50 g of which contain 1 g non volatile solute (molar mass $256\ \text{g mol}^{-1}$ ) and the decrease in freezing point is 0.40 K ?

(A) $4.43\ \text{K kg mol}^{-1}$

(B) $3.72\ \text{K kg mol}^{-1}$

(C) $5.12\ \text{K kg mol}^{-1}$

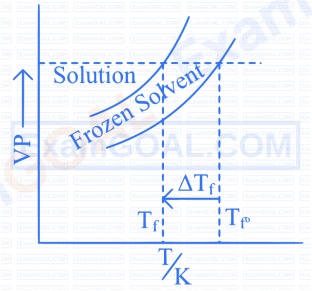
(D) $1.86\ \text{K kg mol}^{-1}$

**Q33.** JEE Main 2025 (Online) 24th January Morning Shift

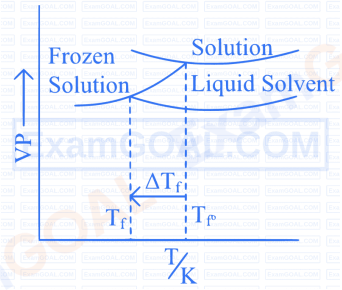
Open solution

Consider the given plots of vapour pressure (VP) vs temperature(T/K). Which amongst the following options is correct graphical representation showing $\Delta T_f$, depression in the freezing point of a solvent in a solution?

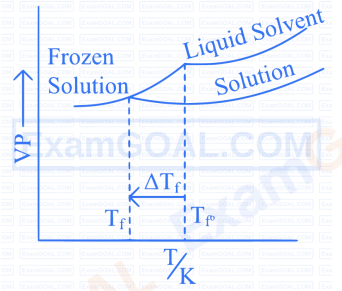
(A)



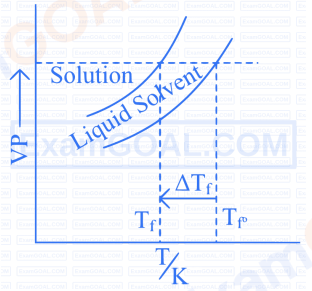

(B)




(C)




(D)




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**Q34.** JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

When a non-volatile solute is added to the solvent, the vapour pressure of the solvent decreases by 10 mm of Hg . The mole fraction of the solute in the solution is 0.2 . What would be the mole fraction of the solvent if decrease in vapour pressure is 20 mm of Hg ?

(A) 0.2 (B) 0.4 (C) 0.8 (D) 0.6

**Q35.** JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Consider a binary solution of two volatile liquid components 1 and 2. $x_1$ and $y_1$ are the mole fractions of component 1 in liquid and vapour phase, respectively. The slope and intercept of the linear plot of $\frac{1}{x_1}$ vs $\frac{1}{y_1}$ are given respectively as :

(A) $\frac{P_1^0}{P_2^0}, \frac{P_1^0-P_2^0}{P_2^0}$

(B) $\frac{P_2^0}{P_1^0}, \frac{P_2^0-P_1^0}{P_2^0}$

(C) $\frac{P_2^0}{P_1^0}, \frac{P_1^0-P_2^0}{P_2^0}$

(D) $\frac{P_1^0}{P_2^0}, \frac{P_2^0-P_1^0}{P_2^0}$

**Q36.** JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Arrange the following solutions in order of their increasing boiling points.

(i) $10^{-4}$MNaCl

(ii) $10^{-4}$M Urea

(iii) $10^{-3}$MNaCl

(iv) $10^{-2}$MNaCl

(A) ( i ) < ( ii ) < ( iii ) < ( iv )

(B) (ii) < ( i ) < ( iii ) < ( iv )

(C) (iv) < ( iii ) < ( i ) < ( ii )

(D) (ii) < (i) ≡ (iii) < (iv)

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**Q37.** JEE Main 2024 (Online) 9th April Morning Shift

Open solution

$0.05\mathrm{M}\ \mathrm{CuSO_4}$ when treated with $0.01\mathrm{M}\ \mathrm{K_2Cr_2O_7}$ gives green colour solution of $\mathrm{Cu_2Cr_2O_7}$. The two solutions are separated as shown below : [SPM : Semi Permeable Membrane]




Due to osmosis :

(A) Molarity of $\mathrm{K_2Cr_2O_7}$ solution is lowered.

(B) Green colour formation observed on side Y.

(C) Molarity of $\mathrm{CuSO_4}$ solution is lowered.

(D) Green colour formation observed on side X.

**Q38.** JEE Main 2024 (Online) 1st February Morning Shift

Open solution

We have three aqueous solutions of $\mathrm{NaCl}$ labelled as ' $\mathrm{A}$ ', ' $\mathrm{B}$ ' and ' $\mathrm{C}$ ' with concentration $0.1\mathrm{M}$, $0.01\mathrm{M}$ and $0.001\mathrm{M}$, respectively. The value of van 't Hoff factor(i) for these solutions will be in the order :

(A) $\mathrm{i_A} < \mathrm{i_C} < \mathrm{i_B}$

(B) $\mathrm{i_A} < \mathrm{i_B} < \mathrm{i_C}$

(C) $\mathrm{i_A} > \mathrm{i_B} > \mathrm{i_C}$

(D) $\mathrm{i_A} = \mathrm{i_B} = \mathrm{i_C}$

**Q39.** JEE Main 2024 (Online) 31st January Morning Shift

Open solution

Identify the mixture that shows positive deviations from Raoult's Law

(A) $(\mathrm{CH_3})_2\mathrm{CO} + \mathrm{CS_2}$

(B) $(\mathrm{CH_3})_2\mathrm{CO} + \mathrm{C_6H_5NH_2}$

(C) $\mathrm{CHCl_3} + \mathrm{C_6H_6}$

(D) $\mathrm{CHCl_3} + (\mathrm{CH_3})_2\mathrm{CO}$

**Q40.** JEE Main 2024 (Online) 30th January Evening Shift

Open solution

The solution from the following with highest depression in freezing point/lowest freezing point is

(A) $180\ \mathrm{g}$ of acetic acid dissolved in benzene

(B) $180\ \mathrm{g}$ of acetic acid dissolved in water

(C) $180\ \mathrm{g}$ of benzoic acid dissolved in benzene

(D) $180\ \mathrm{g}$ of glucose dissolved in water

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**Q41.** JEE Main 2024 (Online) 30th January Morning Shift

What happens to freezing point of benzene when small quantity of napthalene is added to benzene?

(A) Increases
(B) Decreases
(C) Remains unchanged
(D) First decreases and then increases

Open solution

**Q42.** JEE Main 2024 (Online) 27th January Morning Shift

A solution of two miscible liquids showing negative deviation from Raoult's law will have :

(A) increased vapour pressure, increased boiling point
(B) increased vapour pressure, decreased boiling point
(C) decreased vapour pressure, decreased boiling point
(D) decreased vapour pressure, increased boiling point

Open solution

**Q43.** JEE Main 2023 (Online) 11th April Evening Shift

What weight of glucose must be dissolved in $100\ \mathrm{g}$ of water to lower the vapour pressure by $0.20\ \mathrm{mm\ Hg}$ ?

(Assume dilute solution is being formed)

Given : Vapour pressure of pure water is $54.2\ \mathrm{mm\ Hg}$ at room temperature. Molar mass of glucose is $180\ \mathrm{g\ mol^{-1}}$

(A) 3.69 g (B) 2.59 g (C) 3.59 g (D) 4.69 g

Open solution

**Q44.** JEE Main 2023 (Online) 31st January Evening Shift

Evaluate the following statements for their correctness.

A. The elevation in boiling point temperature of water will be same for $0.1\mathrm{M\ NaCl}$ and $0.1\mathrm{M}$ urea.

B. Azeotropic mixtures boil without change in their composition.

C. Osmosis always takes place from hypertonic to hypotonic solution.

D. The density of $32\%\ \mathrm{H_2SO_4}$ solution having molarity $4.09\ \mathrm{M}$ is approximately $1.26\ \mathrm{g\ mL^{-1}}$

E. A negatively charged sol is obtained when KI solution is added to silver nitrate solution.

Choose the correct answer from the options given below :

(A) A, B and D only (B) A and C only (C) B and D only (D) B, D and E only

Open solution

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**Q45.** JEE Main 2023 (Online) 29th January Evening Shift

Open solution

Match List I with List II

| | List I | | List II |
|---|---|---|---|
| A. | van't Hoff factor, i | I. | Cryoscopic constant |
| B. | $k_f$ | II. | Isotonic solutions |
| C. | Solutions with same osmotic pressure | III. | $\frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$ |
| D. | Azeotropes | IV. | Solutions with same composition of vapour above it |

Choose the correct answer from the options given below :

(A) A-III, B-I, C-IV, D-II

(B) A-III, B-I, C-II, D-IV

(C) A-III, B-II, C-I, D-IV

(D) A-I, B-III, C-II, D-IV

**Q46.** JEE Main 2023 (Online) 24th January Morning Shift

Open solution

In the depression of freezing point experiment

A. Vapour pressure of the solution is less than that of pure solvent

B. Vapour pressure of the solution is more than that of pure solvent

C. Only solute molecules solidify at the freezing point

D. Only solvent molecules solidify at the freezing point

Choose the most appropriate answer from the options given below :

(A) A and C only

(B) A only

(C) A and D only

(D) B and C only

**Q47.** JEE Main 2022 (Online) 27th July Morning Shift

Open solution

Boiling point of a $2\%$ aqueous solution of a non-volatile solute A is equal to the boiling point of $8\%$ aqueous solution of a non-volatile solute B. The relation between molecular weights of A and B is

(A) $M_A = 4M_B$

(B) $M_B = 4M_A$

(C) $M_A = 8M_B$

(D) $M_B = 8M_A$

**Q48.** JEE Main 2022 (Online) 25th July Evening Shift

Open solution

Two solutions A and B are prepared by dissolving 1 g of non-volatile solutes X and Y, respectively in 1 kg of water. The ratio of depression in freezing points for A and B is found to be 1 : 4. The ratio of molar masses of X and Y is

(A) 1 : 4

(B) 1 : 0.25

(C) 1 : 0.20

(D) 1 : 5

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**Q49.** JEE Main 2022 (Online) 25th July Morning Shift

Open solution

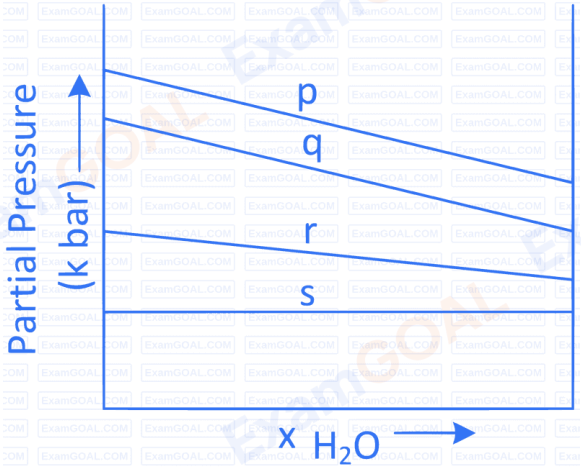
The depression in freezing point observed for a formic acid solution of concentration $0.5\ \text{mL L}^{-1}$ is $0.0405^{\circ}\text{C}$. Density of formic acid is $1.05\ \text{g mL}^{-1}$. The Van't Hoff factor of the formic acid solution is nearly : (Given for water $k_f = 1.86\ \text{k kg mol}^{-1}$ )

(A) 0.8 (B) 1.1 (C) 1.9 (D) 2.4

**Q50.** JEE Main 2022 (Online) 30th June Morning Shift

Open solution

For a solution of the gases A, B, C and D in water at 298 K, the values of Henry's law constant ($K_H$) are 30.40, 2.34, $1.56 \times 10^{-5}$ and 0.513 k bar respectively. In the given graph, the lines marked as 'p' and 's' correspond respectively to :




(A) A and C (B) B and A (C) D and A (D) C and D

**Q51.** JEE Main 2022 (Online) 27th June Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : At 10°C, the density of a 5 M solution of KCl [atomic masses of K & Cl are 39 & 35.5 g $\text{mol}^{-1}$ respectively], is 'x' g $\text{ml}^{-1}$. The solution is cooled to $-21^{\circ}\text{C}$. The molality of the solution will remain unchanged.

Reason (R) : The molality of a solution does not change with temperature as mass remains unaffected with temperature.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

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**Q52.** JEE Main 2022 (Online) 25th June Evening Shift

Solute A associates in water. When 0.7 g of solute A is dissolved in 42.0 g of water, it depresses the freezing point by 0.2°C. The percentage association of solute A in water, is :

[Given : Molar mass of A = 93 g $mol^{-1}$. Molal depression constant of water is 1.86 K kg $mol^{-1}$.]

(A) 50% (B) 60% (C) 70% (D) 80%

Open solution

**Q53.** JEE Main 2021 (Online) 31st August Morning Shift

Which one of the following 0.10 M aqueous solutions will exhibit the largest freezing point depression?

(A) hydrazine (B) glucose (C) glycine (D) $KHSO_4$

Open solution

**Q54.** JEE Main 2021 (Online) 22th July Evening Shift

Which one of the following 0.06 M aqueous solutions has lowest freezing point?

(A) $Al_2(SO_4)_3$ (B) $C_6H_{12}O_6$ (C) KI (D) $K_2SO_4$

Open solution

**Q55.** JEE Main 2020 (Online) 6th September Evening Slot

A set of solutions is prepared using 180 g of water as a solvent and 10 g of different nonvolatile solutes A, B and C. The relative lowering of vapour pressure in the presence of these solutes are in the order :

[Given, molar mass of A = 100 g $mol^{-1}$; B = 200 g $mol^{-1}$; C = 10,000 g $mol^{-1}$]

(A) A > C > B (B) C > B > A (C) A > B > C (D) B > C > A

Open solution

**Q56.** JEE Main 2020 (Online) 2nd September Evening Slot

The size of a raw mango shrinks to a much smaller size when kept in a concentrated salt solution. Which one of the following processes can explain this?

(A) Osmosis (B) Reverse osmosis (C) Diffusion (D) Dialysis

Open solution

**Q57.** JEE Main 2020 (Online) 2nd September Morning Slot

An open beaker of water in equilibrium with water vapour is in a sealed container. When a few grams of glucose are added to the beaker of water, the rate at which water molecules :

(A) leaves the solution increases (B) leaves the vapour increases

(C) leaves the vapour decreases (D) leaves the solution decreases

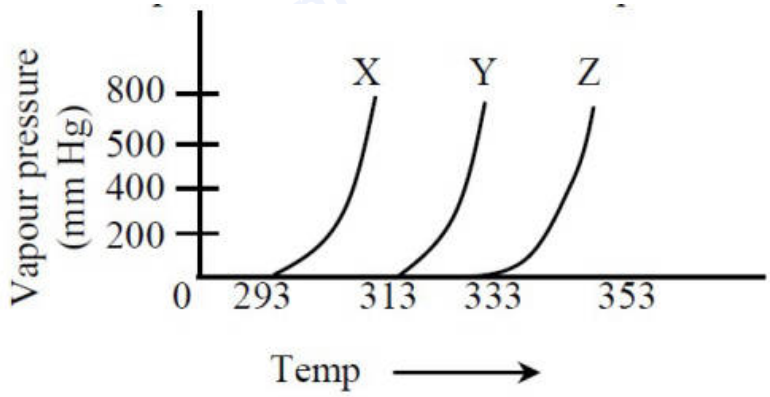
Open solution

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**Q58.** JEE Main 2020 (Online) 8th January Morning Slot

Open solution

A graph of vapour pressure and temperature for three different liquids X, Y, and Z is shown below :




The following inferences are made :
(A) X has higher intermolecular interactions compared to Y.
(B) X has lower intermolecular interactions compared to Y.
(C) Z has lower intermolecular interactions compared to Y.
The correct inference (s) is / are :

**(A)** (B) **(B)** (A) **(C)** (C) **(D)** (A) and (C)

**Q59.** JEE Main 2020 (Online) 7th January Evening Slot

Open solution

Two open beakers one containing a solvent and the other containing a mixture of that solvent with a non volatile solute are together sealed in a container. Over time :

**(A)** The volume of the solution increases and the volume of the solvent decreases.

**(B)** The volume of solution does not change and the volume of the solvent decreases.

**(C)** The volume of the solution and the solvent does not change.

**(D)** The volume of the solution decreases and the volume of the solvent increases.

**Q60.** JEE Main 2020 (Online) 7th January Morning Slot

Open solution

At 35$^{o}$C, the vapour pressure of $CS_2$ is 512 mm. Hg and that of acetone is 344 mm Hg. A solution of $CS_2$ in acetone has a total vapour pressure of 600 mm Hg. The false statement amongst the following is :

**(A)** $CS_2$ and acetone are less attracted to each other than to themselves.

**(B)** a mixture of 100 ml. $CS_2$ and 100 ml. acetone has a volume < 200 ml.

**(C)** Heat must be absorbed in order to produce the solution at 35$^{o}$C

**(D)** Raoult's law is not obeyed by this system.

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**Q61.** JEE Main 2019 (Online) 12th April Evening Slot

A solution is prepared by dissolving 0.6 g of urea (molar mass = 60 g $mol^{-1}$) and 1.8 g of glucose (molar mass = 180 g $mol^{-1}$) in 100 mL of water at 27°C. The osmotic pressure of the solution is : (R = 0.08206 L atm $K^{-1}$ $mol^{-1}$)

Open solution

(A) 8.2 atm (B) 2.46 atm (C) 4.92 atm (D) 1.64 atm

**Q62.** JEE Main 2019 (Online) 10th April Evening Slot

1 g of a non-volatile non-electrolyte solute is dissolved in 100 g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1 : 5. The ratio of the elevation in their boiling points, $\frac{\Delta T_b(A)}{\Delta T_b(B)}$, is :

Open solution

(A) 5 : 1 (B) 1 : 0.2 (C) 10 : 1 (D) 1 : 5

**Q63.** JEE Main 2019 (Online) 10th April Morning Slot

At room temperature, a dilute solution of urea is prepared by dissolving 0.60 g of urea in 360 g of water. If the vapour pressure of pure water at this temperature is 35 mm Hg, lowering of vapour pressure will be. (molar mass of urea = 60 g $mol^{-1}$)

Open solution

(A) 0.031 mmHg (B) 0.017 mmHg (C) 0.028 mmHg (D) 0.027 mmHg

**Q64.** JEE Main 2019 (Online) 9th April Evening Slot

Molal depression constant for a solvent is 4.0 K kg $mol^{-1}$. The depression in the freezing point of the solvent for 0.03 mol $kg^{-1}$ solution of $K_2SO_4$ is :
(Assume complete dissociation of the electrolyte)

Open solution

(A) 0.18 K (B) 0.24 K (C) 0.36 K (D) 0.12 K

**Q65.** JEE Main 2019 (Online) 9th April Morning Slot

The osmotic pressure of a dilute solution of an ionic compound XY in water is four times that of a solution of 0.01 M $BaCl_2$ in water. Assuming complete dissociation of the given ionic compounds in water, the concentration of XY (in mol $L^{-1}$) in solution is :

Open solution

(A) $4 \times 10^{-4}$ (B) $6 \times 10^{-2}$ (C) $4 \times 10^{-2}$ (D) $16 \times 10^{-4}$

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**Q66.** JEE Main 2019 (Online) 9th April Morning Slot

Open solution

Liquid 'M' and liquid 'N' form an ideal solution. The vapour pressures of pure liquids 'M' and 'N' are 450 and 700 mmHg, respectively, at the same temperature. Then correct statement is:
($x_M$ = Mole fraction of 'M' in solution ;
$x_N$ = Mole fraction of 'N' in solution ;
$y_M$ = Mole fraction of 'M' in vapour phase ;
$y_N$ = Mole fraction of 'N' in vapour phase)

(A) $\frac{x_M}{x_N} < \frac{y_M}{y_N}$

(B) $(x_M - y_M) < (x_N - y_N)$

(C) $\frac{x_M}{x_N} = \frac{y_M}{y_N}$

(D) $\frac{x_M}{x_N} > \frac{y_M}{y_N}$

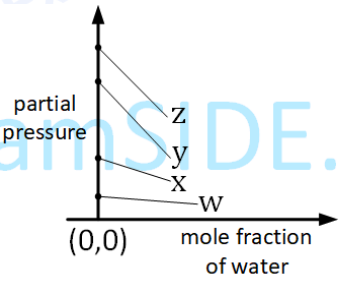
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**Q67.** JEE Main 2019 (Online) 8th April Evening Slot

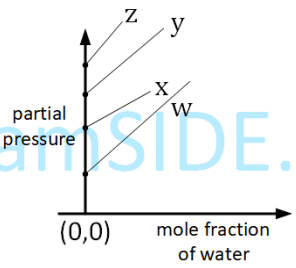
Open solution

For the solution of the gases w, x, y and z in water at 298K, the Henrys law constants ($K_H$) are 0.5, 2, 35 and 40 kbar, respectively. The correct plot for the given data is :-

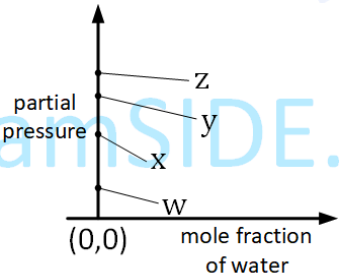
(A)



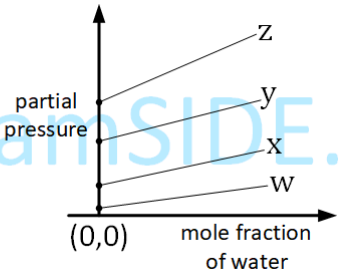

(B)




(C)




(D)




**Q68.** JEE Main 2019 (Online) 8th April Morning Slot

Open solution

The vapour pressures of pure liquids A and B are 400 and 600 mmHg, respectively at 298 K. On mixing the two liquids, the sum of their initial volume is equal to the volume of the final mixture. The mole fraction of liquid B is 0.5 in the mixture, The vapour pressure of the final solution, the mole fractions of components A and B in vapour phase, respectively are :

(A) 500 mmHg, 0.5, 0.5

(B) 500 mmHg, 0.4, 0.6

(C) 450 mmHg, 0.4,0.6

(D) 450 mmHg, 0.5, 0.5

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**Q69.** JEE Main 2019 (Online) 12th January Evening Slot

Molecules of benzoic acid ($C_6H_5COOH$) dimerise in benzene. 'w' g of the acid dissolved in 30 g of benzene shows a depression in freezing point equal to 2K. If the percentage association of the acid to form dimmer in the solution is 80, then w is – (Its given that $K_f$ = 5 K kg $mol^{-1}$, Molar mass of benzoic acid = 122 g $mol^{-1}$)

(A) 1.5 g (B) 1.8 g (C) 1.0 g (D) 2.4 g

Open solution

**Q70.** JEE Main 2019 (Online) 12th January Morning Slot

Freezing point of a 4% aqueous solution of X is equal to freezing point of 12% aqueous solution of Y. If molecular weight of X is A, then molecular weight of Y is -

(A) 4A (B) 2A (C) 3A (D) A

Open solution

**Q71.** JEE Main 2019 (Online) 11th January Evening Slot

$K_2HgI_4$ is 40% ionised in aqueous solution. The value of its van't Hoff factor (i) is:

(A) 1.6 (B) 2.2 (C) 2.0 (D) 1.8

Open solution

**Q72.** JEE Main 2019 (Online) 11th January Morning Slot

The freezing point of a diluted milk sample is found to be $-0.2^oC$, while it should have been $-0.5^oC$ for pure milk. How much water has been added to pure milk to make the diluted sample?

(A) 1 cup of water to 2 cups of pure milk (B) 2 cups of water to 3 cups of pure milk
(C) 3 cups of water to 2 cups of pure milk (D) 1 cup of water to 3 cups of pure milk

Open solution

**Q73.** JEE Main 2019 (Online) 10th January Evening Slot

Elevation in the boiling point for 1 molal solution of glucose is 2 K. The depression in the freezing point for 2 molal solution of glucose in the same solvent is 2 K. The relation between $K_b$ and $K_f$ is

(A) $K_b = K_f$ (B) $K_b = 0.5\ K_f$ (C) $K_b = 1.5\ K_f$ (D) $K_b = 2\ K_f$

Open solution

**Q74.** JEE Main 2019 (Online) 10th January Morning Slot

Liquids A and B form an ideal solution in the entire composition range. At 350 K, the vapour pressures of pure A and pure B are $7 \times 10^3$ Pa and $12 \times 10^3$ Pa, respectively. The composition of the vapor in equilibrium with a solution containing 40 mole percent of A at this temperature is :

(A) $x_A = 0.76; x_B = 0.24$ (B) $x_A = 0.28; x_B = 0.72$
(C) $x_A = 0.4; x_B = 0.6$ (D) $x_A = 0.37; x_B = 0.63$

Open solution

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**Q75.** JEE Main 2019 (Online) 9th January Evening Slot

A solution containing 62 g ethylene glycol in 250 g water is cooled to $-10^{o}C$. If $K_f$ for water is 1.86 K kg $mol^{-1}$, the amount of water (in g) separated as ice is :

(A) 48 (B) 32 (C) 64 (D) 16

Open solution

**Q76.** JEE Main 2019 (Online) 9th January Morning Slot

Which one of the following statements regarding Henry's law is not correct ?

(A) Higher the value of $K_H$ at a given pressure, higher is the solubility of the gas in the liquids

(B) Different gases have different $K_H$ (Henry's law constant) values at the same temperature.

(C) The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution.

(D) The value of $K_H$ increases with increase of temperature and $K_H$ is function of the nature of the gas

Open solution

**Q77.** JEE Main 2018 (Online) 16th April Morning Slot

The mass of a non-volatile, non-electrolyte solute (molar mass = 50 g $mol^{-1}$) needed to be dissolved in 114 g octane to reduce its vapour pressure by 75%, is :

(A) 37.5 g (B) 75 g (C) 150 g (D) 50 g

Open solution

**Q78.** JEE Main 2018 (Offline)

For 1 molal aqueous solution of the following compounds, which one will show the highest freezing point?

(A) $[Co(H_2O)_3\ Cl_3].3H_2O$ (B) $[Co(H_2O)_6]\ Cl_3$

(C) $[Co(H_2O)_5\ Cl]\ Cl_2.H_2O$ (D) $[Co(H_2O)_4\ Cl_2]\ Cl.2H_2O$

Open solution

**Q79.** JEE Main 2018 (Online) 15th April Evening Slot

Two 5 molal solutions are prepared by dissolving a non-electrolyte non-volatile solute separately in the solvents X and Y. The molecular weights of the solvents are $M_x$ and $M_y$, respectively where $M_x = \frac{3}{4} M_y$. The relative lowering of vapor pressure of the solution in X is "m" times that of the solution in Y. Given that the number of moles of solute is very small in comparison to that of the solvent, the value of "m" is :

(A) $\frac{4}{3}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Open solution

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**Q80.** JEE Main 2017 (Online) 9th April Morning Slot

A solution is prepared by mixing 8.5 g of $CH_2Cl_2$ and 11.95 g of $CHCl_3$ . If vapour pressure of $CH_2Cl_2$ and $CHCl_3$ at 298 K are 415 and 200 mmHg respectively, the mole fraction of $CHCl_3$ in vapour form is : (Molar mass of Cl = 35.5 g $mol^{-1}$)

Open solution

(A) 0.162 (B) 0.675 (C) 0.325 (D) 0.486

**Q81.** JEE Main 2017 (Online) 8th April Morning Slot

5 g of $Na_2SO_4$ was dissolved in x g of $H_2O$. The change in freezing point was found to be 3.82°C. If $Na_2SO_4$ is 81.5% ionised, the value of x

($K_f$ for water=1.86°C kg $mol^{-1}$) is approximately :

(molar mass of S = 32 g $mol^{-1}$ and that of Na = 23 g $mol^{-1}$)

Open solution

(A) 15 g (B) 25 g (C) 45 g (D) 65 g

**Q82.** JEE Main 2017 (Offline)

The freezing point of benzene decreases by 0.45°C when 0.2 g of acetic acid is added to 20g of benzene. If acetic acid associates to form a dimer in benzene, percentage association of acetic acid in benzene will be: ($K_f$ for benzene = 5.12 K kg $mol^{-1}$)

Open solution

(A) 80.4 % (B) 74.6 % (C) 94.6 % (D) 64.6 %

**Q83.** JEE Main 2016 (Online) 10th April Morning Slot

An aqueous solution of a salt $MX_2$ at certain temperature has a van't Hoff factor of 2. The degree of dissociation for this solution of the salt is :

Open solution

(A) 0.33 (B) 0.50 (C) 0.67 (D) 0.80

**Q84.** JEE Main 2016 (Online) 9th April Morning Slot

The solubility of $N_2$ in water at 300 K and 500 torr partial pressure is 0.01 g $L^{-1}$. The solubility (in g $L^{-1}$) at 750 torr partial pressure is :

Open solution

(A) 0.0075 (B) 0.015 (C) 0.02 (D) 0.005

**Q85.** JEE Main 2016 (Offline)

18 g glucose ($C_6H_{12}O_6$) is added to 178.2 g water. The vapor pressure of water (in torr) for this aqueous solution is :

Open solution

(A) 76.0 (B) 752.4 (C) 759.0 (D) 7.6

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**Q86.** JEE Main 2015 (Online) 11th April Morning Slot

Determination of the molar mass of acetic acid in benzene using freezing point depression is affected by :

(A) dissociation
(B) association
(C) partial ionization
(D) complex formation

Open solution

**Q87.** JEE Main 2015 (Online) 10th April Morning Slot

A solution at $20^oC$ is composed of 1.5 mol of benzene and 3.5 mol of toluene. If the vapour pressure of pure benzene and pure toluene at this temperature are 74.7 torr and 22.3 torr, respectively, then the total vapour pressure of the solution and the benzene mole fraction in equilibrium with it will be, respectively :

(A) 35.0 torr and 0.480
(B) 38.0 torr and 0.589
(C) 30.5 torr and 0.389
(D) 35.8 torr and 0.280

Open solution

**Q88.** JEE Main 2015 (Offline)

The vapour pressure of acetone at $20^oC$ is 185 torr. When 1.2 g of a non-volatile substance was dissolved in 100 g of acetone at $20^oC$, its vapour pressure was 183 torr. The molar mass (g $mol^{-1}$) of the substance is:

(A) 64
(B) 128
(C) 488
(D) 32

Open solution

**Q89.** JEE Main 2014 (Online) 19th April Morning Slot

Choose the correct statement with respect to the vapour pressure of a liquid among the following :

(A) Increases linearly with increasing temperature
(B) increases non-linearly with increasing temperature
(C) Decreases linearly with increasing temperature
(D) Decreases non-linearly with increasing temperature

Open solution

**Q90.** JEE Main 2014 (Online) 19th April Morning Slot

The observed osmotic pressure for a 0.10 M solution of $Fe(NH_4)_2(SO_4)_2$ at $25^oC$ is 10.8 atm. The expected and experimental (observed) values of Van't Hoff factor (i) will be respectively :
(R = 0.082 L atm $k^{-1}$ $mol^{-1}$)

(A) 5 and 4.42
(B) 4 and 4.00
(C) 5 and 3.42
(D) 3 and 5.42

Open solution

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**Q91.** JEE Main 2014 (Online) 19th April Morning Slot

For an ideal solution of two components A and B, which of the following is true ?

(A) $\Delta H_{mixing} < 0$ (zero)

(B) $\Delta H_{mixing} > 0$ (zero)

(C) A – B interaction is stronger than A – A and B – B interactions

(D) A – A, B – B and A – B interactions are identical

Open solution

**Q92.** JEE Main 2014 (Offline)

Consider separate solutions of 0.500 M $C_2H_5OH(aq)$, 0.100 M $Mg_3(PO_4)_2(aq)$, 0.250 M KBr(aq) and 0.125 M $Na_3PO_4(aq)$ at 25°C. Which statement is true about these solutions, assuming all salts to be strong electrolytes?

(A) 0.125 M $Na_3PO_4(aq)$ has the highest osmotic pressure.

(B) 0.500 M $C_2H_5OH(aq)$ has the highest osmotic pressure

(C) They all have the same osmotic pressure

(D) 0.100 M $Mg_3(PO_4)_2(aq)$ has the highest osmotic pressure.

Open solution

**Q93.** JEE Main 2013 (Online) 25th April Morning Slot

How many grams of methyl alcohol should be added to a 10 litre tank of water to prevent its freezing at 268 K? ($K_f$ for water is 1.86 K kg $mol^{-1}$)

(A) 880.07 g (B) 899.04 g (C) 886.02 g (D) 868.81 g

Open solution

**Q94.** JEE Main 2013 (Online) 23th April Morning Slot

Vapour pressure of pure benzene is 119 torr and that of toluene is 37.0 torr at the same temperature. Mole fraction of toluene in vapour phase which is in equilibrium with a solution of benzene and toluene having a mole fraction of toluene 0.50, will be :

(A) 0.137 (B) 0.237 (C) 0.435 (D) 0.205

Open solution

**Q95.** JEE Main 2013 (Online) 22th April Morning Slot

A molecule M associates in a given solvent according to the equation $M \rightleftharpoons (M)_n$. For a certain concentration of M, the van't Hoff factor was found to be 0.9 and the fraction of associated molecules was 0.2. The value of n is:

(A) 3 (B) 5 (C) 2 (D) 4

Open solution

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**Q96.** JEE Main 2013 (Online) 9th April Morning Slot

12 g of a nonvolatile solute dissolved in 108 g of water produces the relative lowering of vapour pressure of 0.1. The molecular mass of the solute is :

(A) 80 (B) 60 (C) 20 (D) 40

Open solution

**Q97.** AIEEE 2012 (Online) 26th May

The freezing point of a $1.00\ \mathrm{m}$ aqueous solution of HF is found to be $-1.91^{\circ}\mathrm{C}$. The freezing point constant of water, $K_f$ is $1.86\ \mathrm{K\,kg\,mol^{-1}}$. The percentage dissociation of HF at this concentration is

(A) 30% (B) 10% (C) 5.2% (D) 2.7%

Open solution

**Q98.** AIEEE 2012 (Online) 19th May

Liquids A and B form an ideal solution. At $30^{\circ}\mathrm{C}$, the total vapour pressure of a solution containing $1\ \mathrm{mol}$ of A and $2\ \mathrm{mol}$ of B is $250\ \mathrm{mmHg}$. The total vapour pressure becomes $300\ \mathrm{mmHg}$ when 1 more mol of A is added to the first solution. The vapour pressures of pure A and B at the same temperature are

(A) $150, 450\ \mathrm{mmHg}$ (B) $125, 150\ \mathrm{mmHg}$ (C) $450, 150\ \mathrm{mmHg}$ (D) $250, 300\ \mathrm{mmHg}$

Open solution

**Q99.** AIEEE 2012 (Online) 12th May

A solution containing $0.85\ \mathrm{g}$ of $\mathrm{ZnCl_2}$ in $125.0\ \mathrm{g}$ of water freezes at $-0.23^{\circ}\mathrm{C}$. The apparent degree of dissociation of the salt is ($K_f$ for water $= 1.86\ \mathrm{K\,kg\,mol^{-1}}$, atomic mass: $\mathrm{Zn} = 65.3$ and $\mathrm{Cl} = 35.5$)

(A) $1.36\%$ (B) $73.5\%$ (C) $7.35\%$ (D) $2.47\%$

Open solution

**Q100.** AIEEE 2012

$K_f$ for water is 1.86K kg $mol^{-1}$. If your automobile radiator holds 1.0 kg of water, how many grams of ethylene glycol ($C_2H_6O_2$) must you add to get the freezing point of the solution lowered to $-2.8^{o}C$ ?

(A) 72 g (B) 93 g (C) 39 g (D) 27 g

Open solution

**Q101.** AIEEE 2011

Ethylene glycol is used as an antifreeze in a cold climate. Mass of ethylene glycol which should be added to 4 kg of water to prevent it from freezing at $-6^{o}C$ will be : [$K_f$ for water = 1.86 K kg $mol^{-1}$ , and molar mass of ethylene glycol = 62 g $mol^{-1}$ )

(A) 204.30 g (B) 400.00 g (C) 304.60 g (D) 804.32 g

Open solution

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**Q102.** AIEEE 2011

The degree of dissociation ($\alpha$) of a weak electrolyte, $A_xB_y$ is related to van't Hoff factor (i) by the expression :

(A) $\alpha = \frac{i-1}{x+y+1}$ (B) $\alpha = \frac{x+y-1}{i-1}$

(C) $\alpha = \frac{x+y+1}{i-1}$ (D) $\alpha = \frac{i-1}{(x+y-1)}$

Open solution

**Q103.** AIEEE 2010

If sodium sulphate is considered to be completely dissociated into cations and anions in aqueous solution, the change in freezing point of water ($\Delta T_f$), when 0.01 mol of sodium sulphate is dissolved in 1 kg of water, is ($K_f$ = 1.86 K kg $mol^{-1}$)

(A) 0.0372 K (B) 0.0558 K (C) 0.0744 K (D) 0.0186 K

Open solution

**Q104.** AIEEE 2010

On mixing, heptane and octane form an ideal solution. At 373 K, the vapour pressures of the two liquid components (heptane and octane) are 105 kPa and 45 kPa respectively. Vapour pressure of the solution obtained by mixing 25.0g of heptane and 35 g of octane will be (molar mass of heptane = 100 g $mol^{-1}$ and of octane = 114 g $mol^{-1}$)

(A) 72.0 kPa (B) 36.1 kPa (C) 96.2 kPa (D) 144.5 kPa

Open solution

**Q105.** AIEEE 2009

A binary liquid solution is prepared by mixing n-heptane and ethanol. Which one of the following statements is correct regarding the behaviour of the solution ?

(A) The solution formed is an ideal solution

(B) The solution is non-ideal, showing +ve deviation from Raoult's law.

(C) The solution is non-ideal, showing –ve deviation from Raoult's law.

(D) n-heptane shows +ve deviation while ethanol shows –ve deviation from Raoult's law.

Open solution

**Q106.** AIEEE 2009

Two liquids X and Y form an ideal solution. At 300K, vapour pressure of the solution containing 1 mol of X and 3 mol of Y is 550 mm Hg. At the same temperature, if 1 mol of Y is further added to this solution, vapour pressure of the solution increases by 10 mm Hg. Vapour pressure (in mm Hg) of X and Y in their pure states will be, respectively :

(A) 200 and 300 (B) 300 and 400 (C) 400 and 600 (D) 500 and 600

Open solution

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**Q107.** AIEEE 2008

At 80°C, the vapour pressure of pure liquid 'A' is 520 mm Hg and that of pure liquid 'B' is 1000 mm Hg. If a mixture solution of 'A' and 'B' boils at 80°C and 1 atm pressure, the amount of 'A' in the mixture is (1 atm = 760 mm Hg)

(A) 52 mol percent (B) 34 mol percent (C) 48 mol percent (D) 50 mol percent

Open solution

**Q108.** AIEEE 2008

The vapour pressure of water at 20°C is 17.5 mm Hg. If 18 g of glucose ($C_6H_{12}O_6$) is added to 178.2 g of water at 20°C, the vapour pressure of the resulting solution will be

(A) 17.675 mm Hg (B) 15.750 mm Hg (C) 16.500 mm Hg (D) 17.325 mm Hg

Open solution

**Q109.** AIEEE 2007

A mixture of ethyl alcohol and propyl alcohol has a vapour pressure of 290 mm at 300 K. The vapour pressure of propyl alcohol is 200 mm. If the mole fraction of ethyl alcohol is 0.6, its vapour pressure (in mm) at the same temperature will be

(A) 360 (B) 350 (C) 300 (D) 700

Open solution

**Q110.** AIEEE 2007

A 5.25 % solution of a substance is isotonic with a 1.5% solution of urea (molar mass = 60 g $mol^{-1}$) in the same solvent. If the densities of both the solutions are assumed to be equal to 1.0 g $cm^{-3}$, molar mass of the substance will be

(A) 90.0 g $mol^{-1}$ (B) 115.0 g $mol^{-1}$ (C) 105.0 g $mol^{-1}$ (D) 210.0 g $mol^{-1}$

Open solution

**Q111.** AIEEE 2006

18 g of glucose ($C_6H_{12}O_6$) is added to 178.2 g of water. The vapour pressure of water for this aqueous solution at 100°C is

(A) 759.00 Torr (B) 7.60 Torr (C) 76.00 Torr (D) 752.40 Torr

Open solution

**Q112.** AIEEE 2005

If $\alpha$ is the degree of dissociation of $Na_2SO_4$, the vant Hoff's factor (i) used for calculating the molecular mass is :

(A) $1 + \alpha$ (B) $1 + 2\alpha$ (C) $1 - \alpha$ (D) $1 - 2\alpha$

Open solution

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**Q113.** AIEEE 2005

Benzene and toluene form nearly ideal solutions. At 20 $^{o}$C, the vapour pressure of benzene is 75 torr and that of toluene is 22 torr. The partial vapour pressure of benzene at 20 $^{o}$C for a solution containing 78 g of benzene and 46 g of toluene in torr is

(A) 50 (B) 25 (C) 53.5 (D) 37.5

Open solution

**Q114.** AIEEE 2005

Equimolar solutions in the same solvent have

(A) Same boiling point but different freezing point

(B) Same freezing point but different boiling point

(C) Same boiling and same freezing points

(D) Different boiling and different freezing points

Open solution

**Q115.** AIEEE 2004

Which one of the following aqueous solutions will exhibit highest boiling point?

(A) 0.01 M $Na_2SO_4$ (B) 0.015 M glucose (C) 0.015 M urea (D) 0.01 M $KNO_3$

Open solution

**Q116.** AIEEE 2004

For which of the following parameters the structural isomers $C_2H_5OH$ and $CH_3OCH_3$ would be expected to have the same values? (Assume ideal behaviour)

(A) Heat of vaporization

(B) Gaseous densities at the same temperature and pressure

(C) Boiling points

(D) Vapour pressure at the same temperature

Open solution

**Q117.** AIEEE 2004

Which of the following liquid pairs shows a positive deviation from Raoult's law?

(A) Water – hydrochloric acid (B) Acetone – chloroform

(C) Water – nitric acid (D) Benzene – methanol

Open solution

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**Q118.** AIEEE 2004

Which one of the following statements is false?

(A) Raoult's law states that the vapour pressure of a component over a solution is proportional to its mole fraction

(B) Two sucrose solutions of same molality prepared in different solvents will have the same freezing point depression

(C) The correct order of osmotic pressure for 0.01 M aqueous solution of each compound is $BaCl_2$ > KCl > $CH_3COOH$ > sucrose

(D) The osmotic pressure ($\pi$) = MRT, where M is the molarity of the solution

Open solution

**Q119.** AIEEE 2003

If liquids A and B form an ideal solution

(A) the entropy of mixing is zero

(B) the free energy of mixing is zero

(C) the free energy as well as the entropy of mixing are each zero

(D) the enthalpy of mixing is zero

Open solution

**Q120.** AIEEE 2003

In a 0.2 molal aqueous solution of a weak acid HX the degree of ionization is 0.3. Taking $k_f$ for water as 1.85, the freezing point of the solution will be nearest to

(A) -0.360°C (B) -0.260°C (C) +0.480°C (D) -0.480°C

Open solution

**Q121.** AIEEE 2003

A pressure cooker reduces cooking time for food because

(A) boiling point of water involved in cooking is increased

(B) the higher pressure inside the cooker crushes the food material

(C) cooking involves chemical changes helped by a rise in temperature

(D) heat is more evenly distributed in the cooking space

Open solution

**Q122.** AIEEE 2002

Freezing point of an aqueous solution is (-0.186)°C. Elevation of boiling point of the same solution is $K_b$ = 0.512 °C, $K_f$ = 1.86 °C, find the increase in boiling point.

(A) 0.186 °C (B) 0.0512 °C (C) 0.092 °C (D) 0.2732 °C

Open solution

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**Q123.** AIEEE 2002

In a mixture of A and B, components show negative deviation when :

(A) $\Delta V_{mix}$ > 0, $\Delta S_{mix}$ > 0

(B) $\Delta V_{mix}$ = 0, $\Delta S_{mix}$ > 0

(C) A - B interaction is weaker than A - A and B - B interaction

(D) A - B interaction is stronger than A - A and B - B interaction

Open solution

## Numerical 84 questions

**Q124.** JEE Main 2026 (Online) 5th April Evening Shift

20 g hemoglobin in a 1 L aqueous solution $(A)$ at 300 K is separated from pure water by semi permeable membrane. At equilibrium the height of solution in a tube dipped in a solution (A) is found to be 80.0 mm higher than the tube dipped in water.

The molar mass of hemoglobin is ____ $\mathrm{kgmol^{-1}}$. (Nearest integer)

(Given : $\mathrm{g = 10\ m\ s^{-2}, R = 8.3 kPadm^3\ K^{-1}\ mol^{-1}}$, density of solution $= 1000\ \mathrm{kg\ m^{-3}}$ )

Open solution

**Q125.** JEE Main 2026 (Online) 4th April Morning Shift

A non-volatile, non-electrolyte solid solute when dissolved in 40 g of a solvent, the vapour pressure of the solvent decreased from 760 mm Hg to 750 mm Hg . If the same solution boils at 320 K , then the number of moles of the solvent present in the solution is ____ . (Nearest integer)

[Given: boiling point of the pure solvent $= 319.5\ \mathrm{K}$,

$\mathrm{K_b}$ of the solvent $= 0.3\ \mathrm{K\ kg\ mol^{-1}}$ ]

Open solution

**Q126.** JEE Main 2026 (Online) 23rd January Evening Shift

Two liquids A and B form an ideal solution. At 320 K , the vapour pressure of the solution, containing 3 mol of $A$ and 1 mol of $B$ is 500 mm Hg . At the same temperature, if 1 mol of A is further added to this solution, vapour pressure of the solution increases by 20 mm Hg . Vapour pressure (in mm Hg ) of B in pure state is ____ . (Nearest integer)

Open solution

**Q127.** JEE Main 2026 (Online) 21st January Evening Shift

The osmotic pressure of a living cell is 12 atm at 300 K. The strength of sodium chloride solution that is isotonic with the living cell at this temperature is _____ g $L^{-1}$. (Nearest integer)

Given : R = 0.08 L atm $K^{-1}$ $mol^{-1}$

Assume complete dissociation of NaCl

(Given: Molar mass of Na and Cl are 23 and 35.5 g $mol^{-1}$ respectively.)

Open solution

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**Q128.** JEE Main 2026 (Online) 21st January Evening Shift

A substance 'X' (1.5 g) dissolved in 150 g of a solvent 'Y' (molar mass = 300 g mol$^{-1}$) led to an elevation of the boiling point by 0.5 K. The relative lowering in the vapour pressure of the solvent 'Y' is _____ × 10$^{-2}$. (nearest integer)

[Given: $K_b$ of the solvent = 5.0 K kg mol$^{-1}$]

Assume the solution to be dilute and no association or dissociation of X takes place in solution.

Open solution

**Q129.** JEE Main 2025 (Online) 7th April Morning Shift

The percentage dissociation of a salt $(MX_3)$ solution at given temperature (van't Hoff factor $i = 2$) is _____ %(Nearest integer)

Open solution

**Q130.** JEE Main 2025 (Online) 2nd April Evening Shift

When 1 g each of compounds AB and $AB_2$ are dissolved in 15 g of water separately, they increased the boiling point of water by 2.7 K and 1.5 K respectively. The atomic mass of A (in amu) is_____ $\times 10^{-1}$ (Nearest integer)

(Given : Molal boiling point elevation constant is $0.5\ K\ kg\ mol^{-1}$)

Open solution

**Q131.** JEE Main 2025 (Online) 29th January Morning Shift

If $A_2B$ is 30% ionised in an aqueous solution, then the value of van't Hoff factor (i) is _____ × 10$^{-1}$.

Open solution

**Q132.** JEE Main 2025 (Online) 24th January Evening Shift

The observed and normal molar masses of compound $MX_2$ are 65.6 and 164 respectively. The percent degree of ionisation of $MX_2$ is _____%. (Nearest integer)

Open solution

**Q133.** JEE Main 2024 (Online) 9th April Evening Shift

The vapor pressure of pure benzene and methyl benzene at $27^\circ C$ is given as 80 Torr and 24 Torr, respectively. The mole fraction of methyl benzene in vapor phase, in equilibrium with an equimolar mixture of those two liquids (ideal solution) at the same temperature is _____ $\times 10^{-2}$ (nearest integer)

Open solution

**Q134.** JEE Main 2024 (Online) 8th April Morning Shift

A solution containing $10\ g$ of an electrolyte $AB_2$ in $100\ g$ of water boils at $100.52^\circ C$. The degree of ionization of the electrolyte $(\alpha)$ is _____ $\times 10^{-1}$. (nearest integer)

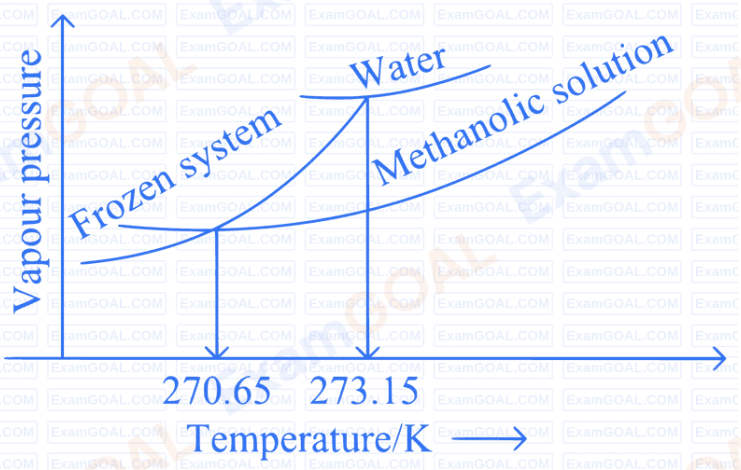
[Given : Molar mass of $AB_2 = 200\ g\ mol^{-1}$, $K_b$ (molal boiling point elevation const. of water) $= 0.52\ K\ kg\ mol^{-1}$, boiling point of water $= 100^\circ C$; $AB_2$ ionises as $AB_2 \rightarrow A^{2+} + 2B^-$]

Open solution

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**Q135.** JEE Main 2024 (Online) 6th April Evening Shift

When '$x$' $\times 10^{-2}$ mL methanol (molar mass $= 32$ g, density $= 0.792$ g/cm$^3$) is added to $100$ mL water (density $= 1$ g/cm$^3$), the following diagram is obtained.




$x =$ _____ (nearest integer).

[Given : Molal freezing point depression constant of water at $273.15$ K is $1.86$ K kg mol$^{-1}$]

Open solution

**Q136.** JEE Main 2024 (Online) 5th April Evening Shift

Considering acetic acid dissociates in water, its dissociation constant is $6.25 \times 10^{-5}$. If $5$ mL of acetic acid is dissolved in 1 litre water, the solution will freeze at $-x \times 10^{-2}{}^{\circ}$C, provided pure water freezes at $0^{\circ}$C.

$x =$ _____. (Nearest integer)

Given : $(K_f)_{water} = 1.86$ K kg mol$^{-1}$

density of acetic acid is $1.2$ g mL$^{-1}$.

molar mass of water $= 18$ g mol$^{-1}$.

molar mass of acetic acid= $60$ g mol$^{-1}$.

density of water $= 1$ g cm$^{-3}$

Acetic acid dissociates as $CH_3COOH \rightleftharpoons CH_3COO^{\ominus} + H^{\oplus}$

Open solution

**Q137.** JEE Main 2024 (Online) 5th April Morning Shift

An artificial cell is made by encapsulating $0.2$ M glucose solution within a semipermeable membrane. The osmotic pressure developed when the artificial cell is placed within a $0.05$ M solution of NaCl at $300$ K is _____ $\times 10^{-1}$ bar. (nearest integer).

[Given : $R = 0.083$ L bar mol$^{-1}$ K$^{-1}$ ]

Assume complete dissociation of NaCl

Open solution

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**Q138.** JEE Main 2024 (Online) 4th April Evening Shift

$2.7\ \text{kg}$ of each of water and acetic acid are mixed. The freezing point of the solution will be $-x^{\circ}\text{C}$. Consider the acetic acid does not dimerise in water, nor dissociates in water. $x =$ _____ (nearest integer)

[Given: Molar mass of water $= 18\ \text{g mol}^{-1}$, acetic acid $= 60\ \text{g mol}^{-1}$

$\text{K}_\text{f}\ \text{H}_2\text{O}: 1.86\ \text{K kg mol}^{-1}$

$\text{K}_\text{f}$ acetic acid: $3.90\ \text{K kg mol}^{-1}$

freezing point: $\text{H}_2\text{O} = 273\ \text{K}$, acetic acid $= 290\ \text{K}$]

Open solution

**Q139.** JEE Main 2024 (Online) 4th April Morning Shift

$2.5\ \text{g}$ of a non-volatile, non-electrolyte is dissolved in $100\ \text{g}$ of water at $25^{\circ}\text{C}$. The solution showed a boiling point elevation by $2^{\circ}\text{C}$. Assuming the solute concentration is negligible with respect to the solvent concentration, the vapor pressure of the resulting aqueous solution is _____ $\text{mm of Hg}$ (nearest integer)

[Given : Molal boiling point elevation constant of water $(\text{K}_\text{b}) = 0.52\ \text{K. kgmol}^{-1}$, $1\ \text{atm}$ pressure $= 760\ \text{mm of Hg}$, molar mass of water $= 18\ \text{g mol}^{-1}$]

Open solution

**Q140.** JEE Main 2024 (Online) 1st February Evening Shift

Mass of ethylene glycol (antifreeze) to be added to $18.6\ \text{kg}$ of water to protect the freezing point at $-24^{\circ}\text{C}$ is _____ $\text{kg}$ (Molar mass in $\text{g mol}^{-1}$ for ethylene glycol $62$, $\text{K}_f$ of water $= 1.86\ \text{K kg mol}^{-1}$ )

Open solution

**Q141.** JEE Main 2024 (Online) 29th January Morning Shift

The osmotic pressure of a dilute solution is $7 \times 10^5\ \text{Pa}$ at $273\ \text{K}$. Osmotic pressure of the same solution at $283\ \text{K}$ is _____ $\times 10^4 \text{Nm}^{-2}$.

Open solution

**Q142.** JEE Main 2023 (Online) 15th April Morning Shift

The vapour pressure of $30\%(\text{w/v})$ aqueous solution of glucose is _____ $\text{mm Hg}$ at $25^{\circ}\text{C}$.

[Given : The density of $30\%$ (w/v), aqueous solution of glucose is $1.2\ \text{g cm}^{-3}$ and vapour pressure of pure water is $24\ \text{mm Hg}$.]

(Molar mass of glucose is $180\ \text{g mol}^{-1}$.)

Open solution

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**Q143.** JEE Main 2023 (Online) 13th April Evening Shift

Sea water contains $29.25\%$ $NaCl$ and $19\%$ $MgCl_2$ by weight of solution. The normal boiling point of the sea water is _____ $^\circ C$ (Nearest integer)

Assume $100\%$ ionization for both $NaCl$ and $MgCl_2$

Given : $K_b\,(H_2O) = 0.52\ K\ kg\ mol^{-1}$

Molar mass of $NaCl$ and $MgCl_2$ is 58.5 and 95 $g\ mol^{-1}$ respectively.

Open solution

**Q144.** JEE Main 2023 (Online) 13th April Morning Shift

Solution of $12\ g$ of non-electrolyte (A) prepared by dissolving it in $1000\ mL$ of water exerts the same osmotic pressure as that of $0.05\ M$ glucose solution at the same temperature. The empirical formula of A is $CH_2O$. The molecular mass of A is _____ g. (Nearest integer)

Open solution

**Q145.** JEE Main 2023 (Online) 12th April Morning Shift

80 mole percent of $MgCl_2$ is dissociated in aqueous solution. The vapour pressure of $1.0$ molal aqueous solution of $MgCl_2$ at $38^\circ C$ is _____ $mm\ Hg$. (Nearest integer)

Given : Vapour pressure of water at $38^\circ C$ is $50\ mm\ Hg$

Open solution

**Q146.** JEE Main 2023 (Online) 11th April Morning Shift

0.004 M $K_2SO_4$ solution is isotonic with 0.01 M glucose solution. Percentage dissociation of $K_2SO_4$ is _____ (Nearest integer)

Open solution

**Q147.** JEE Main 2023 (Online) 10th April Evening Shift

An aqueous solution of volume $300\ cm^3$ contains $0.63\ g$ of protein. The osmotic pressure of the solution at $300\ K$ is 1.29 mbar. The molar mass of the protein is _____ $g\ mol^{-1}$

Given : R = 0.083 L bar $K^{-1}$ $mol^{-1}$

Open solution

**Q148.** JEE Main 2023 (Online) 10th April Morning Shift

If the degree of dissociation of aqueous solution of weak monobasic acid is determined to be 0.3, then the observed freezing point will be _____% higher than the expected/theoretical freezing point. (Nearest integer)

Open solution

**Q149.** JEE Main 2023 (Online) 8th April Evening Shift

If the boiling points of two solvents X and Y (having same molecular weights) are in the ratio $2:1$ and their enthalpy of vaporizations are in the ratio $1:2$, then the boiling point elevation constant of X is $\underline{m}$ times the boiling point elevation constant of Y. The value of m is _____ (nearest integer)

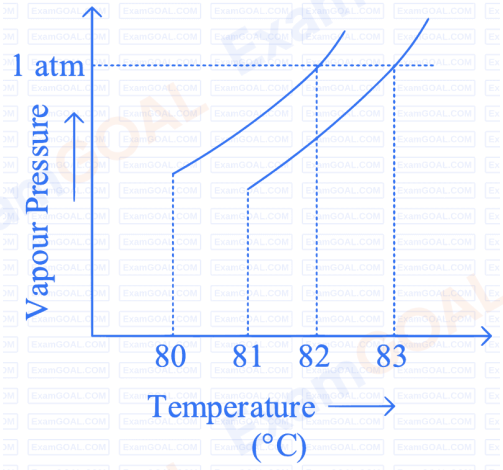
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**Q150.** JEE Main 2023 (Online) 8th April Morning Shift

Open solution

The vapour pressure vs. temperature curve for a solution solvent system is shown below.




The boiling point of the solvent is _____ $^{\circ}\mathrm{C}$.

**Q151.** JEE Main 2023 (Online) 6th April Evening Shift

Open solution

Consider the following pairs of solution which will be isotonic at the same temperature. The number of pairs of solutions is / are _____.

A. $1\ \mathrm{M}$ aq. $\mathrm{NaCl}$ and $2\ \mathrm{M}$ aq. urea

B. $1\ \mathrm{M}$ aq. $\mathrm{CaCl_2}$ and $1.5\ \mathrm{M}$ aq. $\mathrm{KCl}$

C. $1.5\ \mathrm{M}$ aq. $\mathrm{AlCl_3}$ and $2\ \mathrm{M}$ aq. $\mathrm{Na_2SO_4}$

D. $2.5\ \mathrm{M}$ aq. $\mathrm{KCl}$ and $1\ \mathrm{M}$ aq. $\mathrm{Al_2(SO_4)_3}$

**Q152.** JEE Main 2023 (Online) 6th April Morning Shift

Open solution

Mass of Urea $(\mathrm{NH_2CONH_2})$ required to be dissolved in $1000\ \mathrm{g}$ of water in order to reduce the vapour pressure of water by $25\%$ is _____ g. (Nearest integer)

Given: Molar mass of N, C, O and H are $14, 12, 16$ and $1\ \mathrm{g\ mol^{-1}}$ respectively

**Q153.** JEE Main 2023 (Online) 1st February Evening Shift

Open solution

$20\%$ of acetic acid is dissociated when its $5\ \mathrm{g}$ is added to $500\ \mathrm{mL}$ of water. The depression in freezing point of such water is _____ $\times 10^{-3}{}^{\circ}\mathrm{C}$.

Atomic mass of $\mathrm{C}, \mathrm{H}$ and $\mathrm{O}$ are 12,1 and 16 a.m.u. respectively.

[Given : Molal depression constant and density of water are $1.86\ \mathrm{K\ kg\ mol^{-1}}$ and $1\ \mathrm{g\ cm^{-3}}$ respectively.]

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**Q154.** JEE Main 2023 (Online) 1st February Morning Shift

25 mL of an aqueous solution of KCl was found to require 20 mL of 1 M $AgNO_3$ solution when titrated using $K_2CrO_4$ as an indicator. What is the depression in freezing point of KCl solution of the given concentration? _____ (Nearest integer).

(Given : $K_f = 2.0\ K\ kg\ mol^{-1}$)

Assume 1) 100% ionization and 2) density of the aqueous solution as 1 g $mL^{-1}$

Open solution

**Q155.** JEE Main 2023 (Online) 31st January Morning Shift

At $27^\circ C$, a solution containing $2.5\ g$ of solute in $250.0\ mL$ of solution exerts an osmotic pressure of $400\ Pa$. The molar mass of the solute is _____ $g\ mol^{-1}$ (Nearest integer)

(Given : $R = 0.083\ L\ bar\ K^{-1}\ mol^{-1}$)

Open solution

**Q156.** JEE Main 2023 (Online) 30th January Evening Shift

Lead storage battery contains $38\%$ by weight solution of $H_2SO_4$. The van't Hoff factor is $2.67$ at this concentration. The temperature in Kelvin at which the solution in the battery will freeze is _____. (Nearest integer).

Given $K_f = 1.8\ K\ kg\ mol^{-1}$

Open solution

**Q157.** JEE Main 2023 (Online) 30th January Morning Shift

A solution containing $2\ g$ of a non-volatile solute in $20\ g$ of water boils at $373.52\ K$. The molecular mass of the solute is _____ $g\ mol^{-1}$. (Nearest integer)

Given, water boils at $373\ K$, $K_b$ for water $= 0.52\ K\ kg\ mol^{-1}$

Open solution

**Q158.** JEE Main 2023 (Online) 29th January Morning Shift

Solid Lead nitrate is dissolved in 1 litre of water. The solution was found to boil at 100.15°C. When 0.2 mol of NaCl is added to the resulting solution, it was observed that the solution froze at $-0.8^\circ$ C. The solubility product of $PbCl_2$ formed is _____ $\times\ 10^{-6}$ at 298 K. (Nearest integer)

Given : $K_b = 0.5$ K kg $mol^{-1}$ and $K_f = 1.8$ K kg $mol^{-1}$. Assume molality to be equal to molarity in all cases.

Open solution

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**Q159.** JEE Main 2023 (Online) 25th January Evening Shift

Open solution

The number of pairs of the solutions having the same value of the osmotic pressure from the following is _____.

(Assume 100% ionization)

A. 0.500 M $C_2H_5OH$ (aq) and 0.25 M KBr (aq)

B. 0.100 M $K_4[Fe(CN)_6]$ (aq) and 0.100 M $FeSO_4(NH_4)_2SO_4$ (aq)

C. 0.05 M $K_4[Fe(CN)_6]$ (aq) and 0.25 M NaCl (aq)

D. 0.15 M NaCl (aq) and 0.1 M $BaCl_2$ (aq)

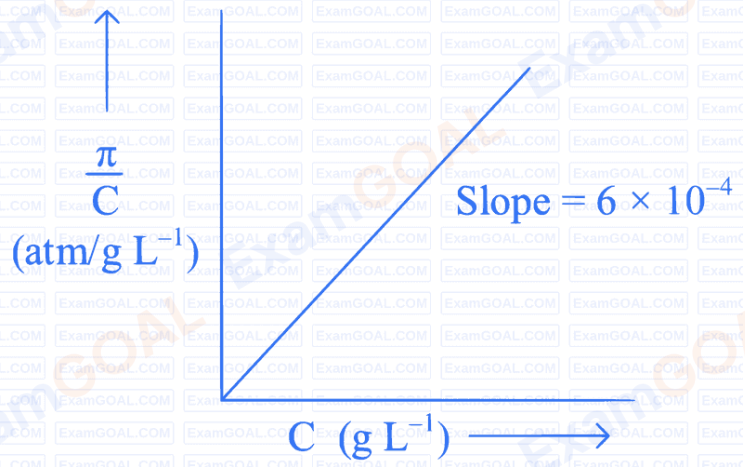
E. 0.02 M KCl. $MgCl_2.6H_2O$ (aq) and 0.05 M KCl (aq)

**Q160.** JEE Main 2023 (Online) 25th January Morning Shift

Open solution

The osmotic pressure of solutions of PVC in cyclohexanone at 300 K are plotted on the graph.

The molar mass of PVC is _____ g $mol^{-1}$ (Nearest integer)




(Given : R = 0.083 L atm $K^{-1}$ $mol^{-1}$)

**Q161.** JEE Main 2023 (Online) 24th January Evening Shift

Open solution

The total pressure observed by mixing two liquids A and B is 350 mm Hg when their mole fractions are 0.7 and 0.3 respectively.

The total pressure becomes 410 mm Hg if the mole fractions are changed to 0.2 and 0.8 respectively for A and B. The vapour pressure of pure A is _____ mm Hg. (Nearest integer)

Consider the liquids and solutions behave ideally.

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**Q162.** JEE Main 2022 (Online) 30th July Evening Shift

31 g of ethylene glycol is mixed with 500 g of water. The freezing point of the aqueous solution is _____ K.

(in nearest integer)

($K_f$ of water $= 1.86\ \mathrm{K\ kg\ mol^{-1}}$)

[Molar mass of C, H, O are 12, 1, 16 $\mathrm{g\ mol^{-1}}$]

Open solution

**Q163.** JEE Main 2022 (Online) 29th July Evening Shift

$1.80\ \mathrm{g}$ of solute A was dissolved in $62.5\ \mathrm{cm^3}$ of ethanol and freezing point of the solution was found to be $155.1\ \mathrm{K}$. The molar mass of solute A is _____ $\mathrm{g\ mol^{-1}}$.

[Given : Freezing point of ethanol is 156.0 K.

Density of ethanol is 0.80 g $\mathrm{cm^{-3}}$.

Freezing point depression constant of ethanol is 2.00 K kg $\mathrm{mol^{-1}}$]

Open solution

**Q164.** JEE Main 2022 (Online) 29th July Morning Shift

If $\mathrm{O_2}$ gas is bubbled through water at $303\ \mathrm{K}$, the number of millimoles of $\mathrm{O_2}$ gas that dissolve in 1 litre of water is _____. (Nearest Integer)

(Given : Henry's Law constant for $\mathrm{O_2}$ at $303\ \mathrm{K}$ is $46.82\ \mathrm{k}$ bar and partial pressure of $\mathrm{O_2} = 0.920$ bar)

(Assume solubility of $\mathrm{O_2}$ in water is too small, nearly negligible)

Open solution

**Q165.** JEE Main 2022 (Online) 28th July Evening Shift

A gaseous mixture of two substances A and B, under a total pressure of $0.8$ atm is in equilibrium with an ideal liquid solution. The mole fraction of substance A is $0.5$ in the vapour phase and $0.2$ in the liquid phase. The vapour pressure of pure liquid $\mathrm{A}$ is _____ atm. (Nearest integer)

Open solution

**Q166.** JEE Main 2022 (Online) 28th July Morning Shift

$150\ \mathrm{g}$ of acetic acid was contaminated with $10.2\ \mathrm{g}$ ascorbic acid $(\mathrm{C_6H_8O_6})$ to lower down its freezing point by $\left(x \times 10^{-1}\right)^{\circ}\mathrm{C}$. The value of $x$ is _____. (Nearest integer)

[Given $\mathrm{K}_f = 3.9\ \mathrm{K\ kg\ mol^{-1}}$; molar mass of ascorbic acid $= 176\ \mathrm{g\ mol^{-1}}$]

Open solution

**Q167.** JEE Main 2022 (Online) 27th July Evening Shift

When a certain amount of solid A is dissolved in $100\ \mathrm{g}$ of water at $25^{\circ}\mathrm{C}$ to make a dilute solution, the vapour pressure of the solution is reduced to one-half of that of pure water. The vapour pressure of pure water is $23.76\ \mathrm{mmHg}$. The number of moles of solute A added is _____. (Nearest Integer)

Open solution

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**Q168.** JEE Main 2022 (Online) 26th July Evening Shift

The elevation in boiling point for 1 molal solution of non-volatile solute A is $3\,\mathrm{K}$. The depression in freezing point for 2 molal solution of $\mathrm{A}$ in the same solvent is 6 $K$. The ratio of $K_b$ and $K_f$ i.e., $K_b/K_f$ is $1 : X$. The value of $X$ is [nearest integer]

Open solution

**Q169.** JEE Main 2022 (Online) 29th June Evening Shift

Elevation in boiling point for 1.5 molal solution of glucose in water is 4 K. The depression in freezing point for 4.5 molal solution of glucose in water is 4 K. The ratio of molal elevation constant to molal depression constant ($K_b/K_f$) is _____.

Open solution

**Q170.** JEE Main 2022 (Online) 29th June Morning Shift

1.2 mL of acetic acid is dissolved in water to make 2.0 L of solution. The depression in freezing point observed for this strength of acid is 0.0198°C. The percentage of dissociation of the acid is _____. (Nearest integer)

[Given : Density of acetic acid is 1.02 g $mL^{-1}$, Molar mass of acetic acid is 60 g $mol^{-1}$, $K_f(H_2O)$ = 1.85 K kg $mol^{-1}$]

Open solution

**Q171.** JEE Main 2022 (Online) 28th June Evening Shift

2.5 g of protein containing only glycine ($C_2H_5NO_2$) is dissolved in water to make 500 mL of solution. The osmotic pressure of this solution at 300 K is found to be $5.03 \times 10^{-3}$ bar. The total number of glycine units present in the protein is _____.

(Given : R = 0.083 L bar $K^{-1}$ $mol^{-1}$)

Open solution

**Q172.** JEE Main 2022 (Online) 28th June Morning Shift

The vapour pressures of two volatile liquids A and B at 25°C are 50 Torr and 100 Torr, respectively. If the liquid mixture contains 0.3 mole fraction of A, then the mole fraction of liquid B in the vapour phase is $\frac{x}{17}$. The value of x is _____.

Open solution

**Q173.** JEE Main 2022 (Online) 27th June Evening Shift

A solution containing $2.5 \times 10^{-3}$ kg of a solute dissolved in $75 \times 10^{-3}$ kg of water boils at 373.535 K. The molar mass of the solute is _____ g $mol^{-1}$. [nearest integer] (Given : $K_b(H_2O)$ = 0.52 K kg $mol^{-1}$ and boiling point of water = 373.15 K)

Open solution

**Q174.** JEE Main 2022 (Online) 27th June Morning Shift

2 g of a non-volatile non-electrolyte solute is dissolved in 200 g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1 : 8. The elevation in boiling points of A and B are in the ratio $\frac{x}{y}$ (x : y). The value of y is _____. (Nearest integer)

Open solution

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**Q175.** JEE Main 2022 (Online) 26th June Evening Shift

The osmotic pressure exerted by a solution prepared by dissolving 2.0 g of protein of molar mass 60 kg $mol^{-1}$ in 200 mL of water at 27°C is _____ Pa. [integer value]

(use R = 0.083 L bar $mol^{-1}$ $K^{-1}$)

Open solution

**Q176.** JEE Main 2022 (Online) 26th June Morning Shift

A 0.5 percent solution of potassium chloride was found to freeze at −0.24°C. The percentage dissociation of potassium chloride is _____. (Nearest integer)

(Molal depression constant for water is 1.80 K kg $mol^{-1}$ and molar mass of KCl is 74.6 g $mol^{-1}$)

Open solution

**Q177.** JEE Main 2022 (Online) 24th June Evening Shift

A company dissolves 'x' amount of $CO_2$ at 298 K in 1 litre of water to prepare soda water. X = _____ $\times 10^{-3}$ g. (nearest integer)

(Given : partial pressure of $CO_2$ at 298 K = 0.835 bar.

Henry's law constant for $CO_2$ at 298 K = 1.67 kbar.

Atomic mass of H, C and O is 1, 12, and 16 g $mol^{-1}$, respectively)

Open solution

**Q178.** JEE Main 2022 (Online) 24th June Morning Shift

The osmotic pressure of blood is 7.47 bar at 300 K. To inject glucose to a patient intravenously, it has to be isotonic with blood. The concentration of glucose solution in $gL^{-1}$ is _____.

(Molar mass of glucose = 180 g $mol^{-1}$, R = 0.083 L bar $K^{-1}$ $mol^{-1}$) (Nearest integer)

Open solution

**Q179.** JEE Main 2021 (Online) 31st August Evening Shift

1.22 g of an organic acid is separately dissolved in 100 g of benzene ($K_b$ = 2.6 K kg $mol^{-1}$) and 100 g of acetone ($K_b$ = 1.7 K kg $mol^{-1}$). The acid is known to dimerize in benzene but remain as a monomer in acetone. The boiling point of the solution in acetone increases by 0.17°C. The increase in boiling point of solution in benzene in °C is $x \times 10^{-2}$. The value of x is _____. (Nearest integer) [Atomic mass : C = 12.0, H = 1.0, O =16.0]

Open solution

**Q180.** JEE Main 2021 (Online) 27th August Evening Shift

40 g of glucose (Molar mass = 180) is mixed with 200 mL of water. The freezing point of solution is _____ K. (Nearest integer) [Given : $K_f$ = 1.86 K kg $mol^{-1}$; Density of water = 1.00 g $cm^{-3}$; Freezing point of water = 273.15 K]

Open solution

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**Q181.** JEE Main 2021 (Online) 27th August Morning Shift

1 kg of 0.75 molal aqueous solution of sucrose can be cooled up to $-4^{\circ}C$ before freezing. The amount of ice (in g) that will be separated out is _____. (Nearest integer)

[Given : $K_f(H_2O) = 1.86$ K kg $mol^{-1}$]

Open solution

**Q182.** JEE Main 2021 (Online) 26th August Evening Shift

83 g of ethylene glycol dissolved in 625 g of water. The freezing point of the solution is _____ K. (Nearest integer)

[Use : Molal Freezing point depression constant of water = 1.86 K kg $mol^{-1}$

Freezing Point of water = 273 K

Atomic masses : C : 12.0 u, O : 16.0 u, H : 1.0 u]

Open solution

**Q183.** JEE Main 2021 (Online) 26th August Morning Shift

Of the following four aqueous solutions, total number of those solutions whose freezing point is lower than that of 0.10 M $C_2H_5OH$ is _____ (Integer answer)

(i) 0.10 M $Ba_3(PO_4)_2$

(ii) 0.10 M $Na_2SO_4$

(iii) 0.10 M KCl

(iv) 0.10 M $Li_3PO_4$

Open solution

**Q184.** JEE Main 2021 (Online) 27th July Evening Shift

In a solvent 50% of an acid HA dimerizes and the rest dissociates. The van't Hoff factor of the acid is _____ $\times 10^{-2}$.

(Round off to the nearest integer)

Open solution

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**Q185.** JEE Main 2021 (Online) 27th July Morning Shift

1.46 g of a biopolymer dissolved in a 100 mL water at 300 K exerted an osmotic pressure of $2.42 \times 10^{-3}$ bar.

The molar mass of the biopolymer is _____ $\times 10^4$ g $mol^{-1}$. (Round off to the Nearest Integer)

[Use : R = 0.083 L bar $mol^{-1}$ $K^{-1}$]

Open solution

**Q186.** JEE Main 2021 (Online) 25th July Evening Shift

When 3.00 g of a substance 'X' is dissolved in 100 g of $CCl_4$, it raises the boiling point by 0.60 K. The molar mass of the substance 'X' is _____ g $mol^{-1}$. (Nearest integer).

[Given $K_b$ for $CCl_4$ is 5.0 K kg $mol^{-1}$]

Open solution

**Q187.** JEE Main 2021 (Online) 25th July Morning Shift

$CO_2$ gas is bubbled through water during a soft drink manufacturing process at 298 K. If $CO_2$ exerts a partial pressure of 0.835 bar then x m mol of $CO_2$ would dissolve in 0.9 L of water. The value of x is _____. (Nearest integer)

(Henry's law constant for $CO_2$ at 298 K is $1.67 \times 10^3$ bar)

Open solution

**Q188.** JEE Main 2021 (Online) 20th July Evening Shift

The vapour pressures of A and B at 25°C are 90 mm Hg and 15 mm Hg respectively. If A and B are mixed such that the mole fraction of A in the mixture is 0.6, then the mole fraction of B in the vapour phase is $x \times 10^{-1}$. The value of x is _____. (Nearest integer)

Open solution

**Q189.** JEE Main 2021 (Online) 20th July Morning Shift

At 20° C, the vapour pressure of benzene is 70 torr and that of methyl benzene is 20 torr. The mole fraction of benzene in the vapour phase at 20°C above an equimolar mixture of benzene and methyl benzene is _____ $\times 10^{-2}$. (Nearest integer)

Open solution

**Q190.** JEE Main 2021 (Online) 18th March Evening Shift

A solute A dimerizes in water. The boiling point of a 2 molal solution of A is 100.52°C. The percentage association of A is _____. (Round off to the Nearest Integer).

[Use : $K_b$ for water = 0.52 K kg $mol^{-1}$ Boiling point of water = 100°C]

Open solution

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**Q191.** JEE Main 2021 (Online) 18th March Morning Shift

2 molal solution of a weak acid HA has a freezing point of $-3.885^{\circ}C$. The degree of dissociation of this acid is _____ $\times 10^{-3}$. (Round off to the Nearest Integer).

[Given : Molal depression constant of water = 1.85 K kg $mol^{-1}$, Freezing point of pure water = $0^{\circ}$ C]

Open solution

**Q192.** JEE Main 2021 (Online) 17th March Evening Shift

A 1 molal $K_4Fe(CN)_6$ solution has a degree of dissociation of 0.4. Its boiling point is equal to that of another solution which contains 18.1 weight percent of a non electrolytic solute A. The molar mass of A is _____ u. (Round off to the Nearest Integer). [Density of water = 1.0 g $cm^{-3}$ ]

Open solution

**Q193.** JEE Main 2021 (Online) 17th March Morning Shift

The oxygen dissolved in water exerts a partial pressure of 20 kPa in the vapour above water. The molar solubility of oxygen in water is _____ $\times 10^{-5}$ mol $dm^{-3}$. (Round off to the Nearest Integer).

[Given : Henry's law constant = $K_H = 8.0 \times 10^4$ kPa for $O_2$. Density of water with dissolved oxygen = 1.0 kg $dm^{-3}$ ]

Open solution

**Q194.** JEE Main 2021 (Online) 16th March Evening Shift

At 363 K, the vapour pressure of A is 21 kPa and that of B is 18 kPa. One mole of A and 2 moles of B are mixed. Assuming that this solution is ideal, the vapour pressure of the mixture is _____ kPa. (Round off to the Nearest Integer).

Open solution

**Q195.** JEE Main 2021 (Online) 16th March Morning Shift

$AB_2$ is 10% dissociated in water to $A^{2+}$ and $B^-$. The boiling point of a 10.0 molal aqueous solution of $AB_2$ is _____$^{\circ}$C. (Round off to the Nearest Integer).

[Given : Molal elevation constant of water $K_b$ = 0.5 K kg $mol^{-1}$, boiling point of pure water = $100^{\circ}$C]

Open solution

**Q196.** JEE Main 2021 (Online) 26th February Evening Shift

When 12.2 g of benzoic acid is dissolved in 100 g of water, the freezing point of solution was found to be $-0.93^{\circ}$C ($K_f(H_2O)$ = 1.86 K kg $mol^{-1}$). The number (n) of benzoic acid molecules associated (assuming 100% association) is _____.

Open solution

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**Q197.** JEE Main 2021 (Online) 26th February Morning Shift

224 mL of $SO_{2(g)}$ at 298 K and 1 atm is passed through 100 mL of 0.1 M NaOH solution. The non-volatile solute produced is dissolved in 36g of water. The lowering of vapour pressure of solution (assuming the solution is dilute) ($P^{o}_{(H_2O)}$ = 24 mm of Hg) is x $\times$ $10^{-2}$ mm of Hg, the value of x is _____. (Integer answer)

Open solution

**Q198.** JEE Main 2021 (Online) 25th February Evening Shift

If a compound AB dissociates to the extent of 75% in an aqueous solution, the molality of the solution which shows a 2.5 K rise in the boiling point of the solution is _____ molal. (Rounded off to the nearest integer)

[$K_b$ = 0.52 K kg $mol^{-1}$]

Open solution

**Q199.** JEE Main 2021 (Online) 25th February Morning Shift

1 molal aqueous solution of an electrolyte $A_2B_3$ is 60% ionised. The boiling point of the solution at 1 atm is _____ K. (Rounded off to the nearest integer)

[Given $K_b$ for ($H_2O$) = 0.52 K kg $mol^{-1}$]

Open solution

**Q200.** JEE Main 2021 (Online) 24th February Evening Shift

$C_6H_6$ freezes at 5.5°C. The temperature at which a solution of 10g of $C_4H_{10}$ in 200g of $C_6H_6$ freeze is _____ °C. (The molal freezing point depression constant of $C_6H_6$ is 5.12°C/m.)

Open solution

**Q201.** JEE Main 2021 (Online) 24th February Morning Shift

When 9.45 g of $ClCH_2COOH$ is added to 500 mL of water, its freezing point drops by 0.5°C. The dissociation constant of $ClCH_2COOH$ is x $\times$ $10^{-3}$.
The value of x is _____. (Rounded off to the nearest integer)
[$K_{f(H_20)}$ = 1.86 K kg $mol^{-1}$]

Open solution

**Q202.** JEE Main 2020 (Online) 6th September Morning Slot

The elevation of boiling point of 0.10 m aqueous $CrCl_3.xNH_3$ solution is two times that of 0.05 m aqueous $CaCl_2$ solution. The value of x is _____.

[Assume 100% ionisation of the complex and $CaCl_2$, coordination number of Cr as 6, and that all $NH_3$ molecules are present inside the coordination sphere]

Open solution

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**Q203.** JEE Main 2020 (Online) 4th September Evening Slot

The osmotic pressure of a solution of NaCl is 0.10 atm and that of a glucose solution is 0.20 atm. The osmotic pressure of a solution formed by mixing 1 L of the sodium chloride solution with 2 L of the glucose solution is $x \times 10^{-3}$ atm. x is _____. (nearest integer)

Open solution

**Q204.** JEE Main 2020 (Online) 4th September Morning Slot

At 300 K, the vapour pressure of a solution containing 1 mole of n-hexane and 3 moles of n-heptane is 550 mm of Hg. At the same temperature, if one more mole of n-heptane is added to this solution, the vapour pressure of the solution increases by 10 mm of Hg. What is the vapour pressure in mm Hg of n-heptane in its pure state _____?

Open solution

**Q205.** JEE Main 2020 (Online) 3rd September Evening Slot

If 250 $cm^3$ of an aqueous solution containing 0.73 g of a protein A is isotonic with one litre of another aqueous solution containing 1.65 g of a protein B, at 298 K, the ratio of the molecular masses of A and B is _____ $\times 10^{-2}$ (to the nearest integer).

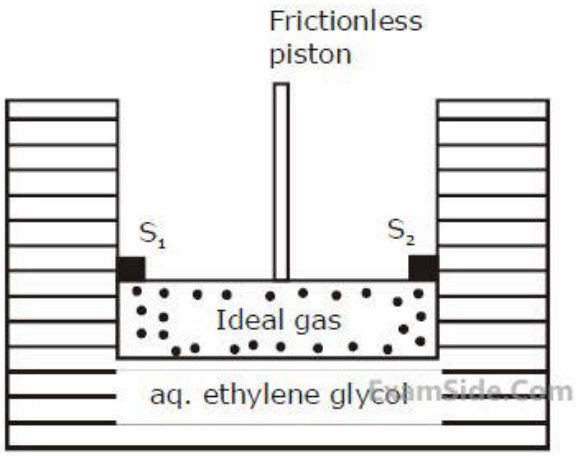
Open solution

**Q206.** JEE Main 2020 (Online) 9th January Evening Slot

A cylinder containing an ideal gas (0.1 mol of 1.0 $dm^3$) is in thermal equilibrium with a large volume of 0.5 molal aqueous solution of ethylene glycol at its freezing point. If the stoppers $S_1$ and $S_2$ (as shown in the figure) are suddenly withdrawn, the volume of the gas in litres after equilibrium is achieved will be____.

(Given, $K_f$ (water) = 2.0 K kg $mol^{-1}$,
R = 0.08 $dm^3$ atm $K^{-1}$ $mol^{-1}$)

Open solution




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**Q207.** JEE Main 2020 (Online) 9th January Morning Slot

How much amount of NaCl should be added to 600 g of water ($\rho$ = 1.00 g/mL) to decrease the freezing point of water to – 0.2 °C ? _____.

(The freezing point depression constant for water = 2K kg $mol^{-1}$)

Open solution

## MCQ (More than One Correct Answer) 1 questions

**Q208.** JEE Main 2020 (Online) 3rd September Morning Slot

Henry's constant (in kbar) for four gases $\alpha$, $\beta$, $\gamma$ and $\delta$ in water at 298 K is given below :

| | $\alpha$ | $\beta$ | $\gamma$ | $\delta$ |
|---|---|---|---|---|
| $K_H$ | 50 | 2 | $2 \times 10^{-5}$ | 0.5 |

(density of water = $10^3$ kg $m^{-3}$ at 298 K)

This table implies that :

(A) solubility of $\gamma$ at 308 K is lower than at 298 K

(B) The pressure of a 55.5 molal solution of $\delta$ is 250 bar

(C) $\alpha$ has the highest solubility in water at a given pressure

(D) The pressure of a 55.5 molal solution of $\gamma$ is 1 bar

Open solution

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# Answer Key

## MCQ (Single Correct Answer)

| | | | | |
|---|---|---|---|---|
| 1. **A** | 2. **C** | 3. **D** | 4. **D** | 5. **C** |
| 6. **A** | 7. **D** | 8. **D** | 9. **C** | 10. **A** |
| 11. **A** | 12. **A** | 13. **D** | 14. **A** | 15. **C** |
| 16. **B** | 17. **A** | 18. **B** | 19. **B** | 20. **D** |
| 21. **A** | 22. **A** | 23. **A** | 24. **A** | 25. **C** |
| 26. **D** | 27. **C** | 28. **B** | 29. **C** | 30. **D** |
| 31. *Scan QR* | 32. **C** | 33. **C** | 34. **D** | 35. **D** |
| 36. **B** | 37. **C** | 38. **B** | 39. **A** | 40. **B** |
| 41. **B** | 42. **D** | 43. **A** | 44. **C** | 45. **B** |
| 46. **C** | 47. **B** | 48. **B** | 49. **C** | 50. **A** |
| 51. **A** | 52. **D** | 53. **D** | 54. **A** | 55. **C** |
| 56. **A** | 57. **D** | 58. **A** | 59. **A** | 60. **B** |
| 61. **C** | 62. **D** | 63. **B** | 64. **C** | 65. **B** |
| 66. **D** | 67. **A** | 68. **B** | 69. **D** | 70. **C** |
| 71. **D** | 72. **C** | 73. **D** | 74. **B** | 75. **C** |
| 76. **A** | 77. **C** | 78. **A** | 79. **B** | 80. **C** |
| 81. **C** | 82. **C** | 83. **B** | 84. **B** | 85. **B** |
| 86. **B** | 87. **B** | 88. **A** | 89. **B** | 90. **A** |
| 91. **D** | 92. **C** | 93. **C** | 94. **B** | 95. **C** |
| 96. **C** | 97. **D** | 98. **C** | 99. **B** | 100. **B** |
| 101. **D** | 102. **D** | 103. **B** | 104. **A** | 105. **B** |
| 106. **C** | 107. **D** | 108. **D** | 109. **B** | 110. **D** |
| 111. **D** | 112. **B** | 113. **A** | 114. **C** | 115. **A** |
| 116. **B** | 117. **D** | 118. **B** | 119. **D** | 120. **D** |
| 121. **A** | 122. **B** | 123. **D** | | |

## Numerical

| | | | | |
|---|---|---|---|---|
| 124. **62** | 125. **5** | 126. **200** | 127. **15** | 128. **3** |
| 129. **33** | 130. **25** | 131. **16** | 132. **75** | 133. **23** |
| 134. **5** | 135. **543** | 136. **19** | 137. **25** | 138. **31** |
| 139. **707** | 140. **15** | 141. **73** | 142. **23** | 143. **116** |

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| | | | | |
|---|---|---|---|---|
| 144. **240** | 145. **48** | 146. **75** | 147. **40535** | 148. **30** |
| 149. **8** | 150. **82** | 151. **4** | 152. **1111** | 153. **372** |
| 154. **3** | 155. **62250** | 156. **243** | 157. **100** | 158. **13** |
| 159. **4** | 160. **41500** | 161. **314** | 162. **271** | 163. **80** |
| 164. **1** | 165. **2** | 166. **15** | 167. **6** | 168. **1** |
| 169. **3** | 170. **5** | 171. **330** | 172. **14** | 173. **45** |
| 174. **8** | 175. **415** | 176. **98** | 177. **1221** | 178. **54** |
| 179. **13** | 180. **271** | 181. **518** | 182. **269** | 183. **4** |
| 184. **125** | 185. **15** | 186. **250** | 187. **25** | 188. **1** |
| 189. **78** | 190. **100** | 191. **50** | 192. **85** | 193. **25** |
| 194. **19** | 195. **106** | 196. **2** | 197. **18 to 24** | 198. **3** |
| 199. **375** | 200. **1** | 201. **34** | 202. **5** | 203. **167** |
| 204. **600** | 205. **177** | 206. **2.17 to 2.23** | 207. **1.74 to 1.76** | |

## MCQ (More than One Correct Answer)

| | |
|---|---|
| 208. **A, B** | |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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